accenture

Learning and Talent Transformation

Azure DevOps

DotNet Case Study Guide

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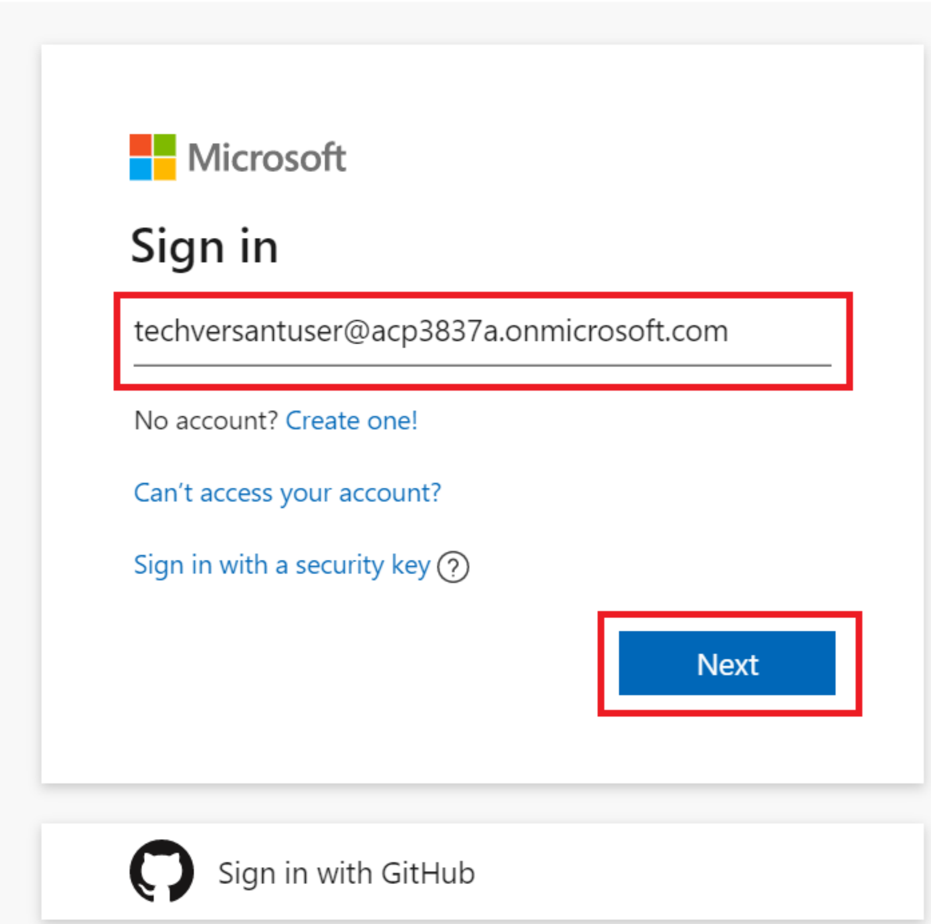
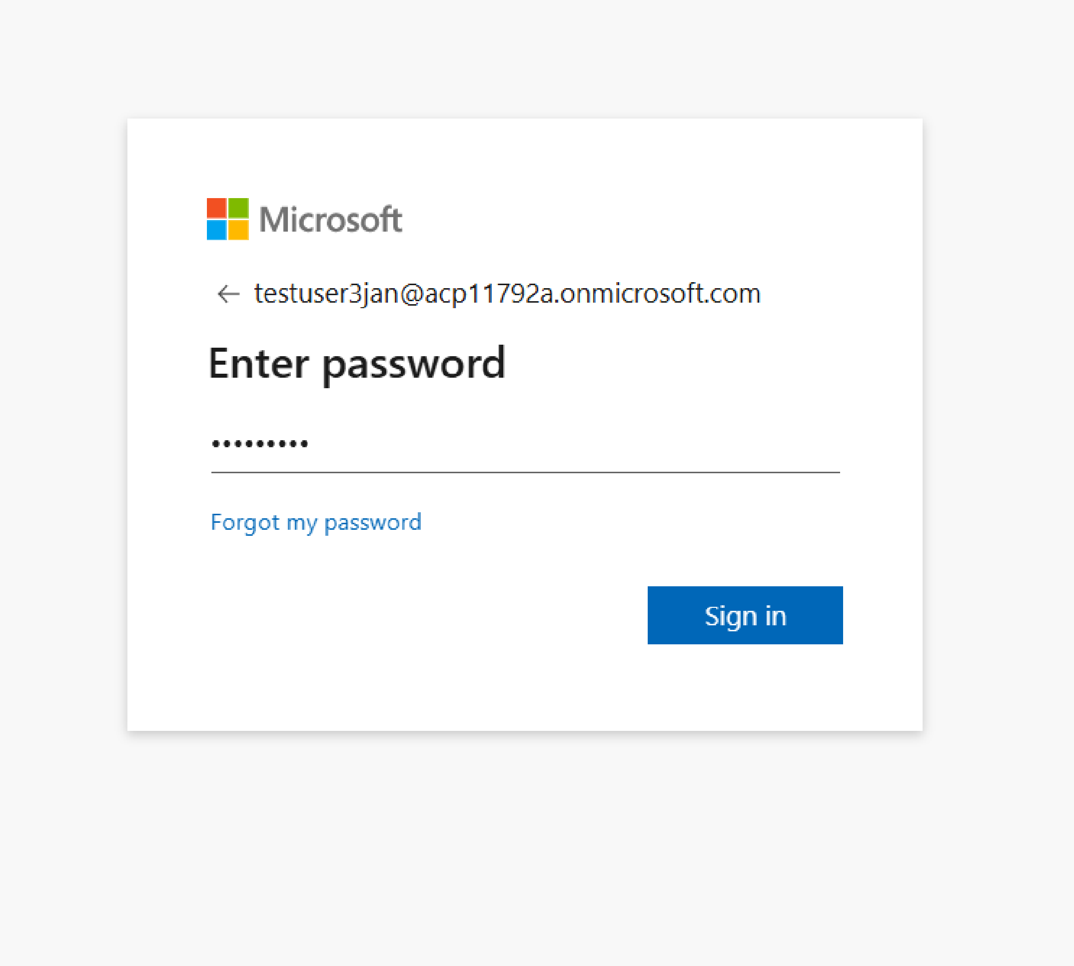
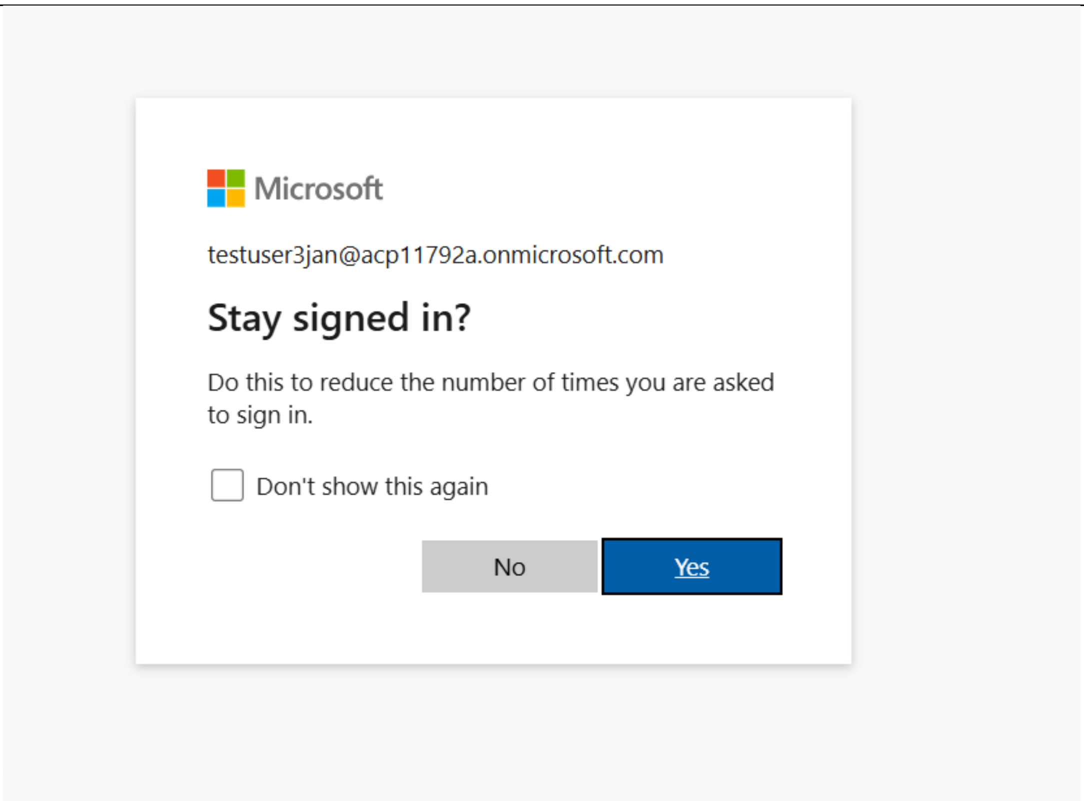
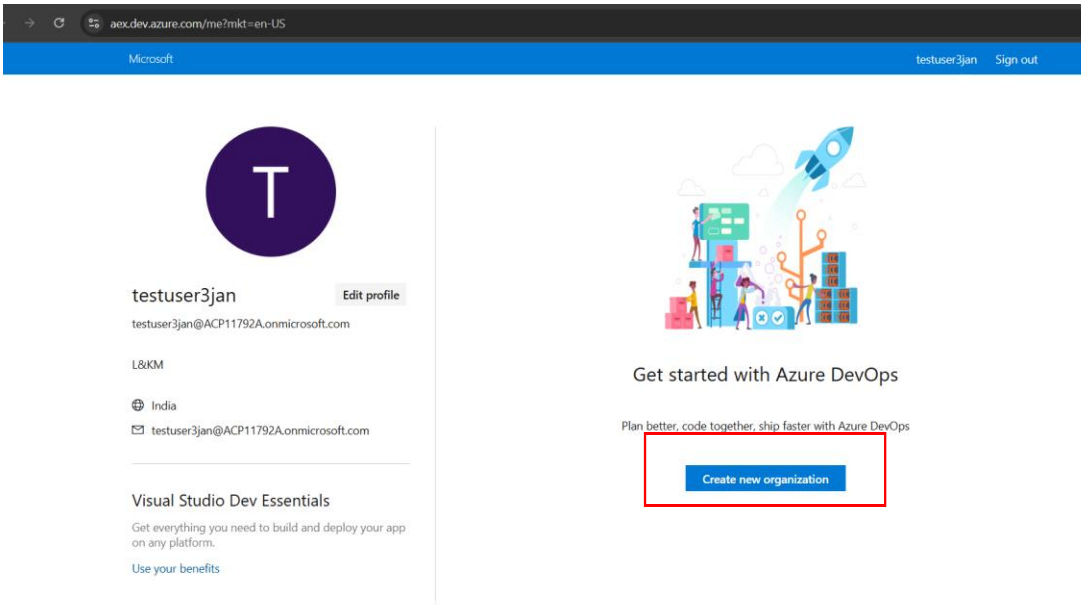
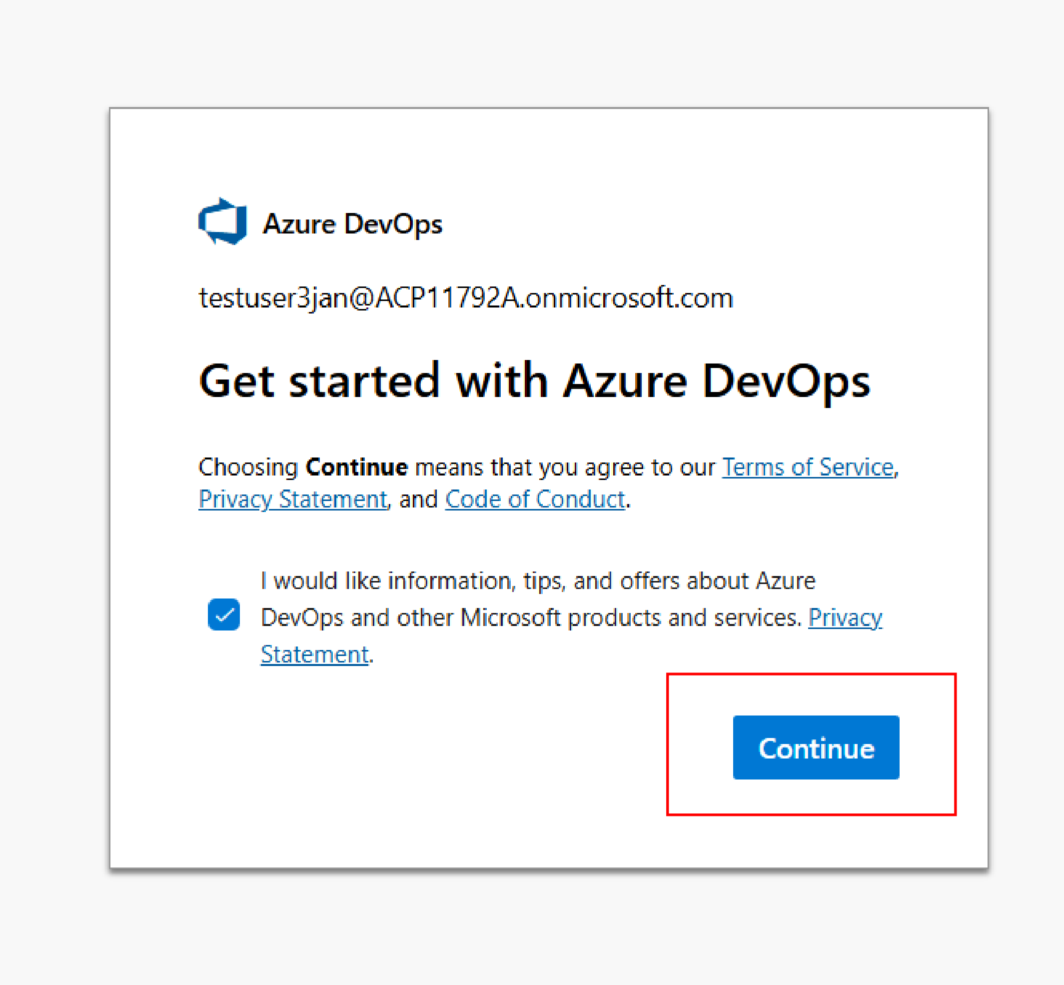
1 AzureDevOps Portal Login

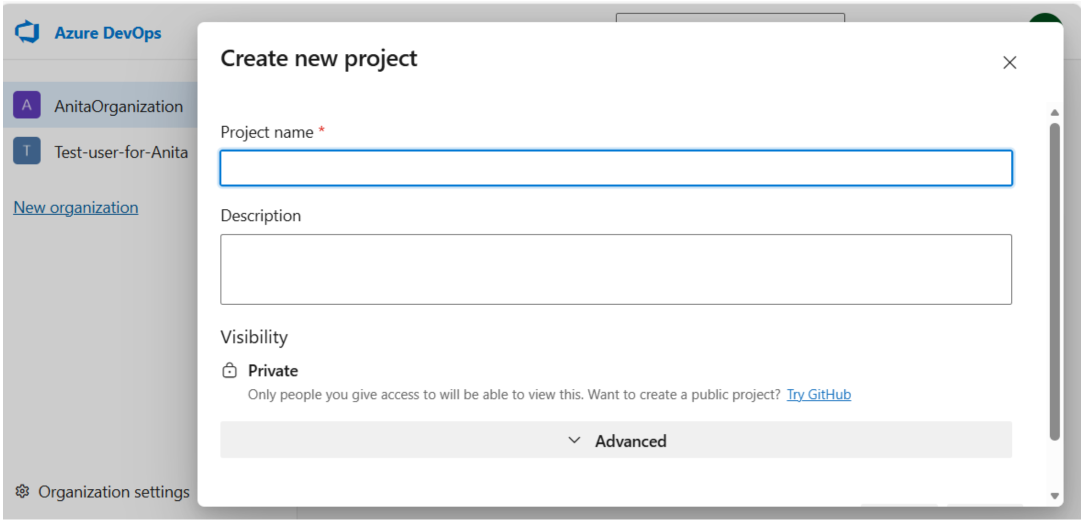
Azure DevOps Portal Login

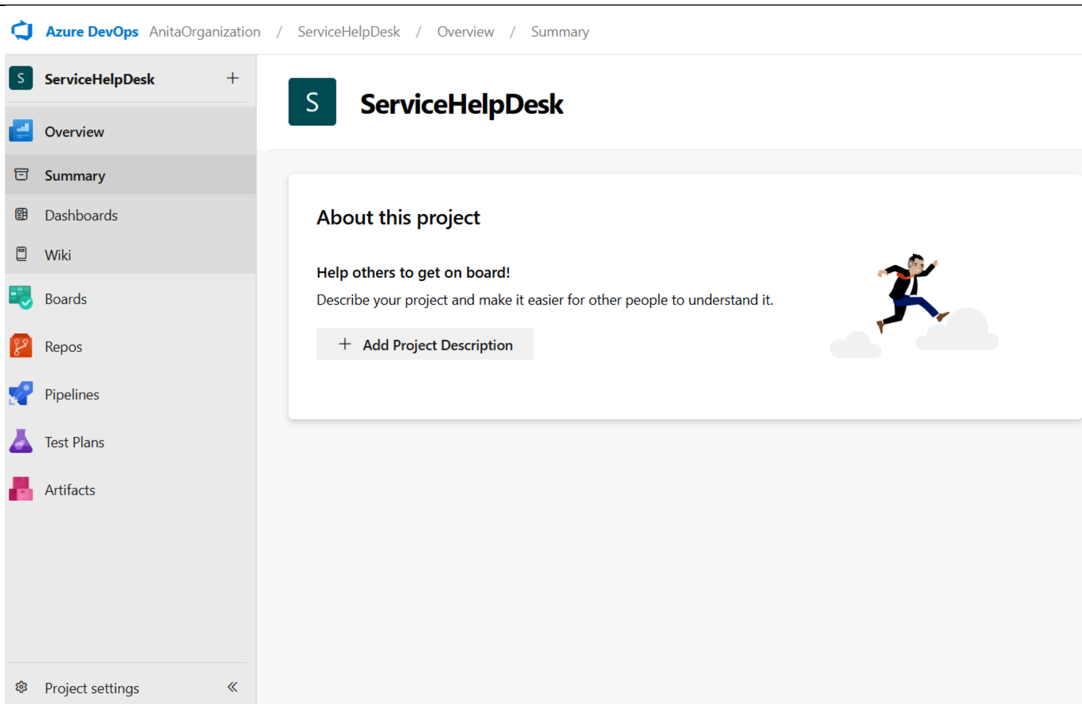
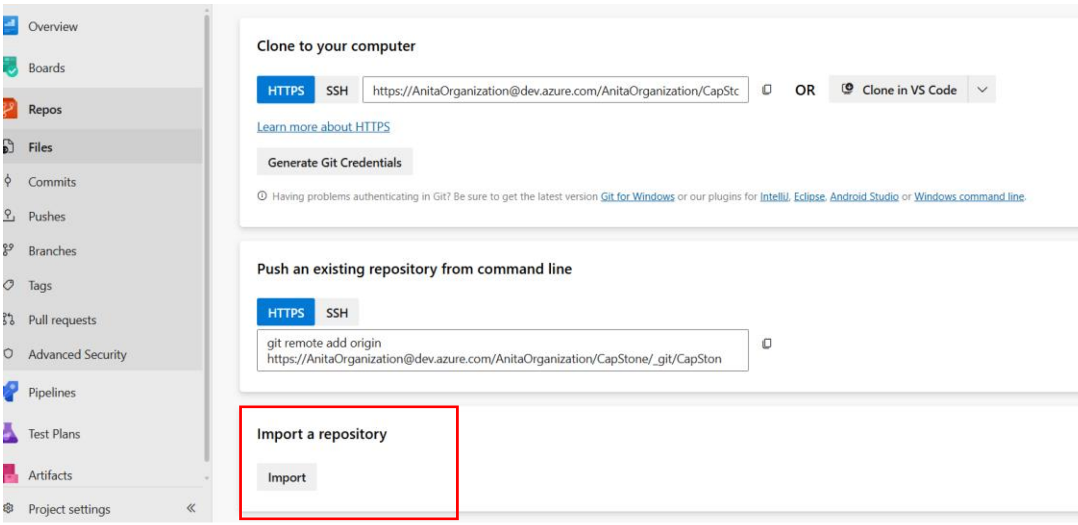
1	Open the chrome browser in incognito mode and navigate to URL https://aex.dev.azure.com/
2	Enter the username provided to you in the login screen. Example username here is "**testuserjan3@acp3837a.onmicrosoft.com**" and click on **Next** button. 
3	Enter the password **Acc1234$$** in the next window.

	
4	Click Next and Complete the MFA through authenticator app.
5	In next window you can click yes or no as per your wish.

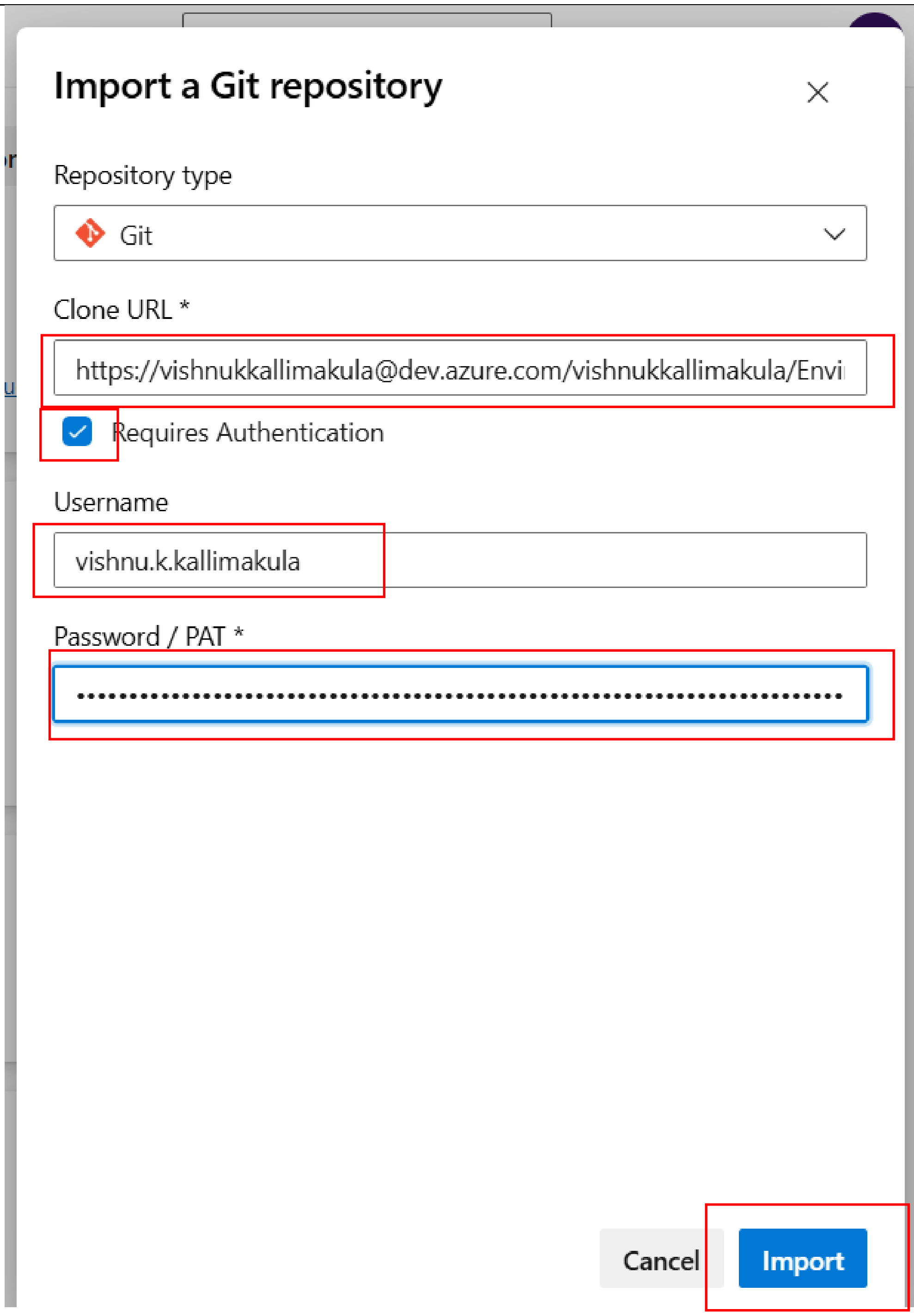
	 After clicking on Yes you will be navigated to the below page, Click on **Create New organization** 

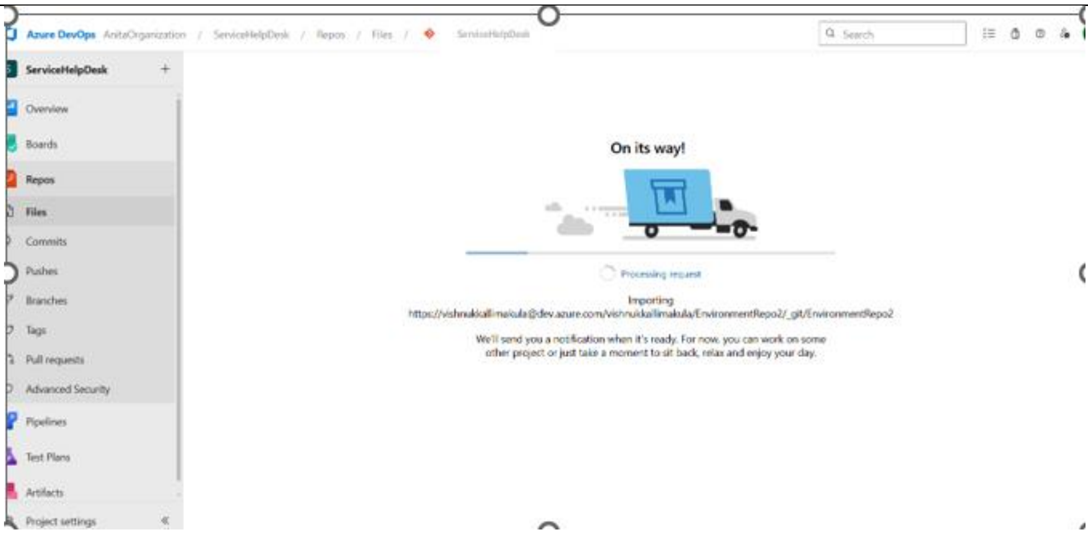
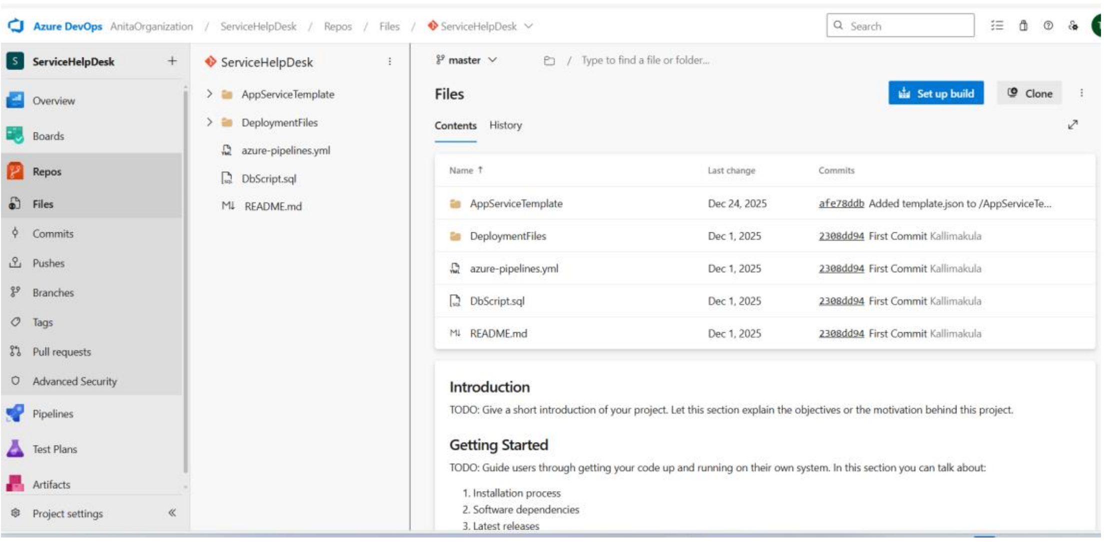
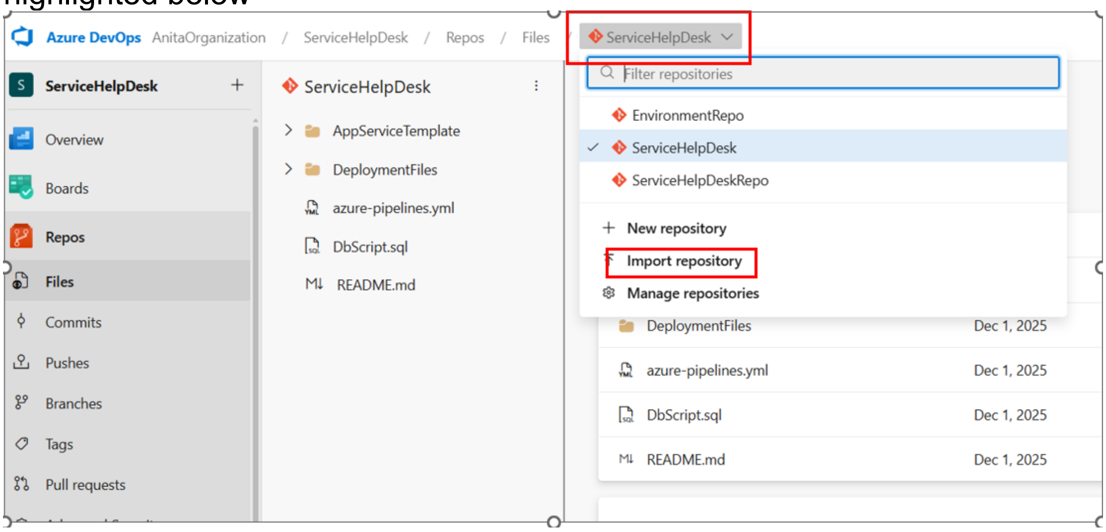
	Now click on **Continue** 
6	In the page enter the organization name as <yourname>Organization. Example here entered as **AnitaOrganization**. Then select location as India and enter the captcha if it ask for it.
7	Once click on Continue. Observe the organization is created and highlighted/selected in the left menu. If not selected , select the appropriate organization you have created.

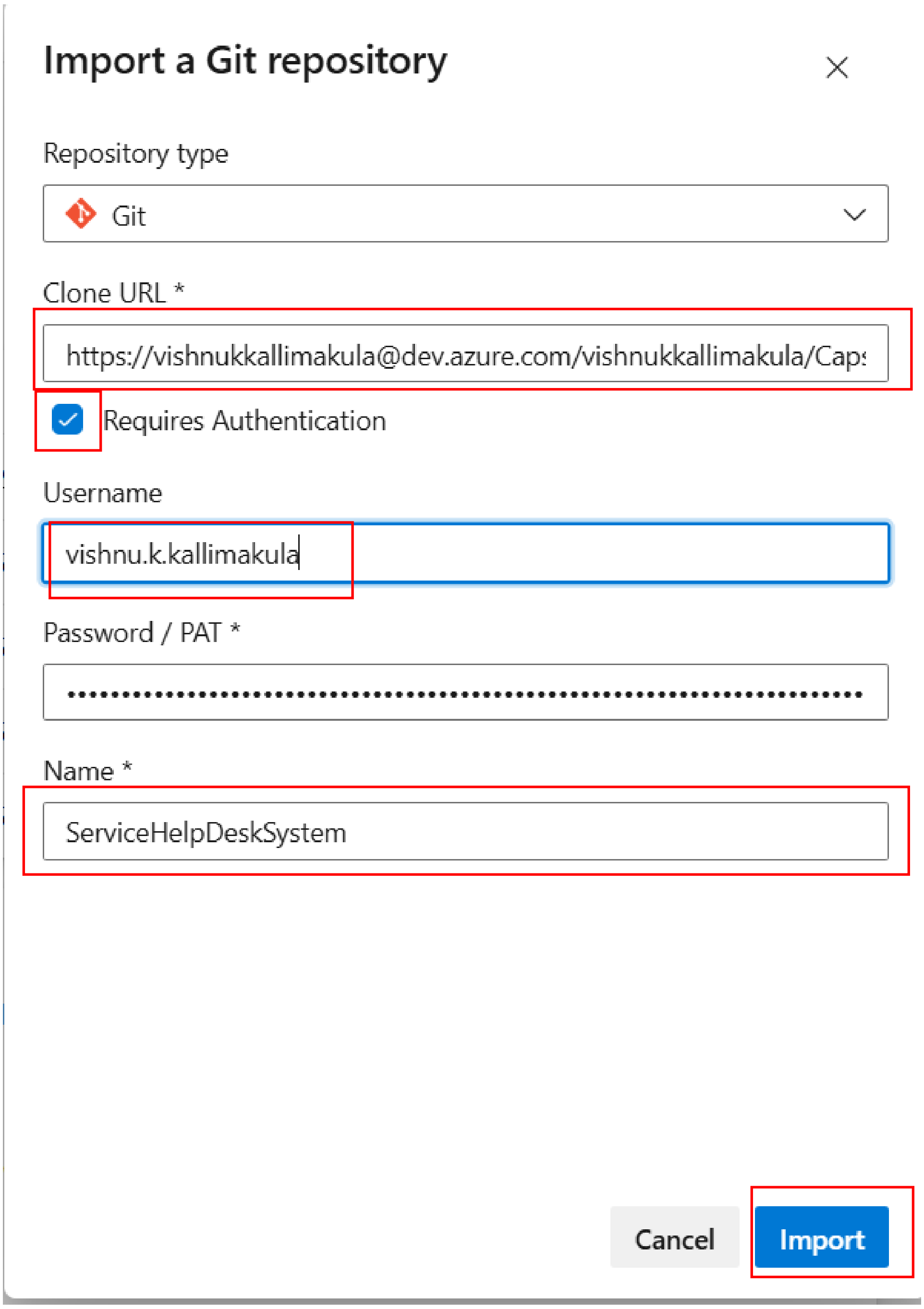
	 
11	Enter the project name as “ServiceHelpDesk” and select visibility as Private and click on Create project button.
12	Now the project overview page is displayed as shown below

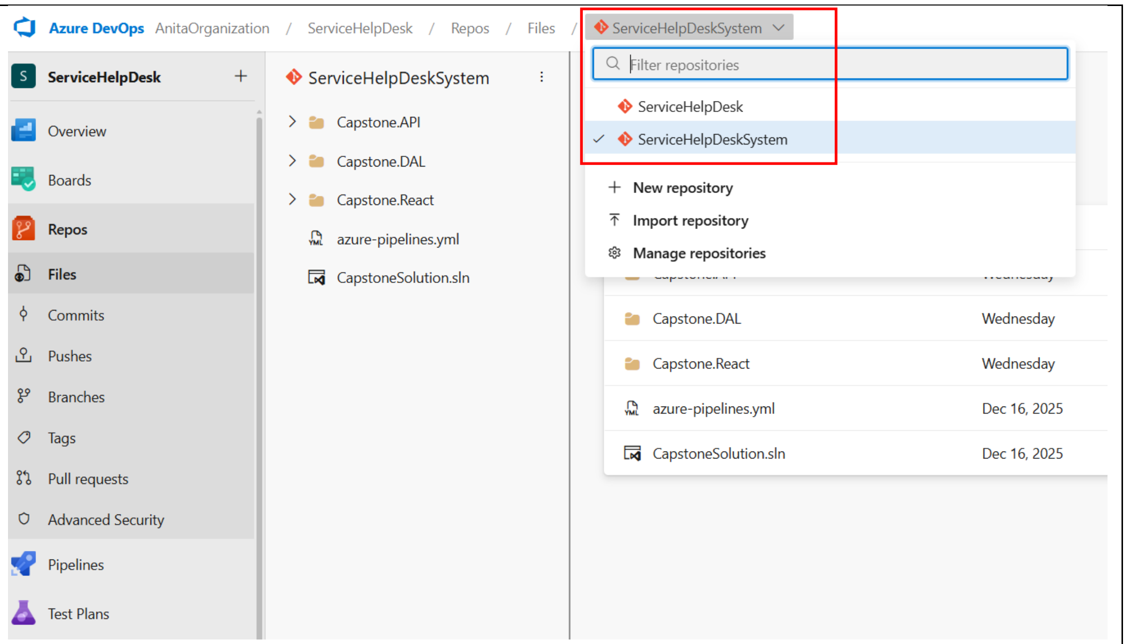
<table>
<tr><td></td><td>
</td></tr>
<tr><td>13</td><td>Now click on Repos from left side and in the right side page displayed, select Import button.

</td></tr>
<tr><td>14</td><td>Select Repository type : Git. Enter the clone URL as
https://vishnukkallimakula@dev.azure.com/vishnukkallimakula/EnvironmentRepo2/_git/EnvironmentRepo2

And click on Import</td></tr>
</table>

	
15	Wait for few seconds for the import to complete.

	 
16	In the similar way import one more repository. Now click project name ServceHelpDesk and select import repository as highlighted below 

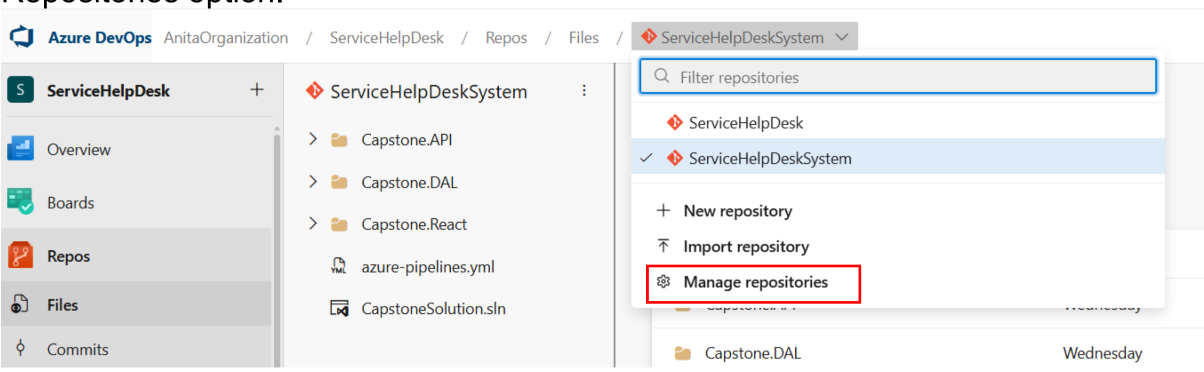
17	Select repo type as Git and enter the clone URL as https://vishnukkallimakula@dev.azure.com/vishnukkallimakula/CapstoneProject1/_git/CapstoneProject1 and click Import 
18	Observe there are two repositories in your project

<table>
<tr><td></td><td>


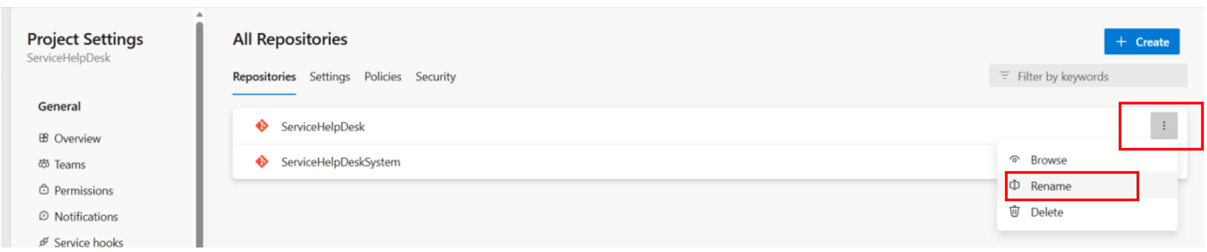
For example here ServiceHelpDeskSystem repo is selected and hence tick mark is present.
</td></tr>
</table>

Rename Repo

<table>
<tr><td>1</td><td>
Rename one repo to EnvironmentRepo to avoid confusion. For that now click on Repos from left side menu and click on Repositories drop down and select Manage Repositories option.

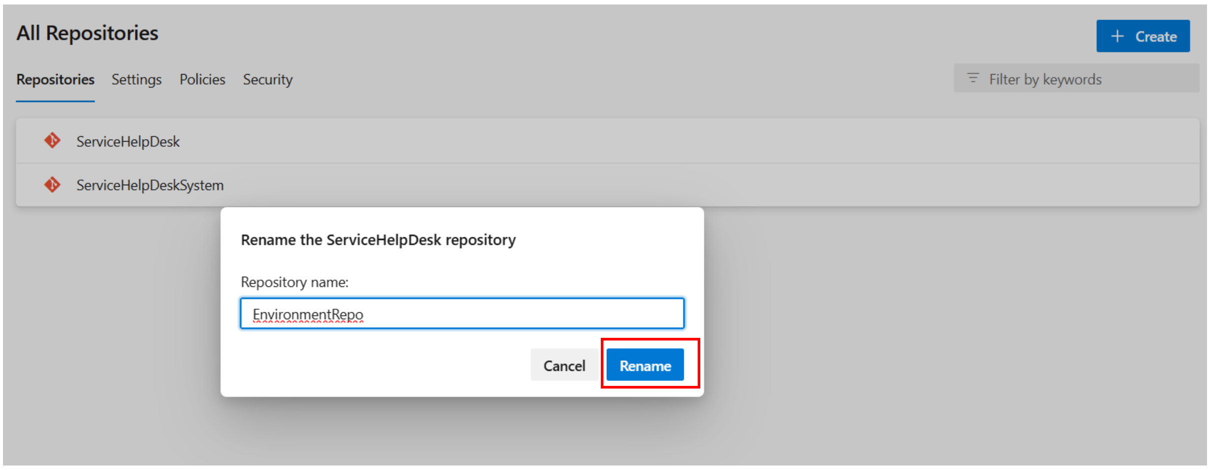


</td></tr>
<tr><td>2</td><td>
Select three dots on the ServiceHelpDesk row and click on rename option

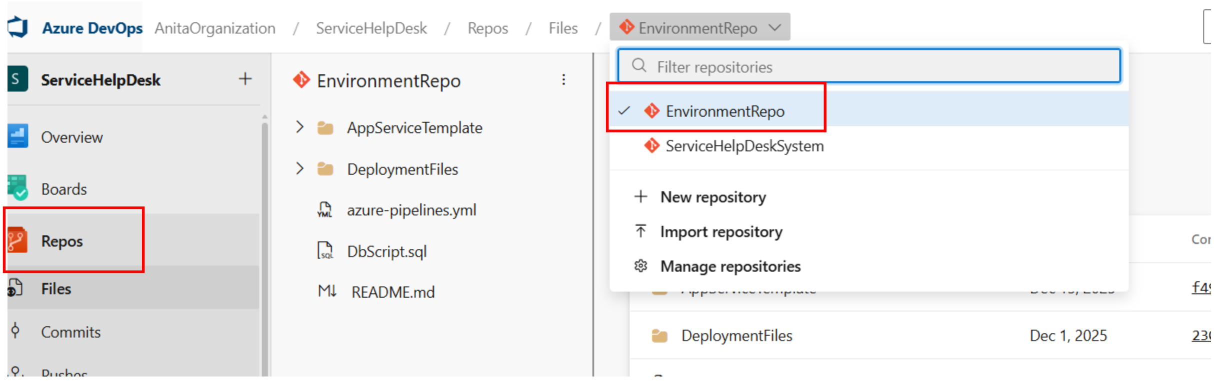


</td></tr>
</table>

<table>
<tr>
<td>3</td>
<td>
Delete the old name **ServiceHelpDesk** and give new name as EnvironmentRepo and click on Rename button.

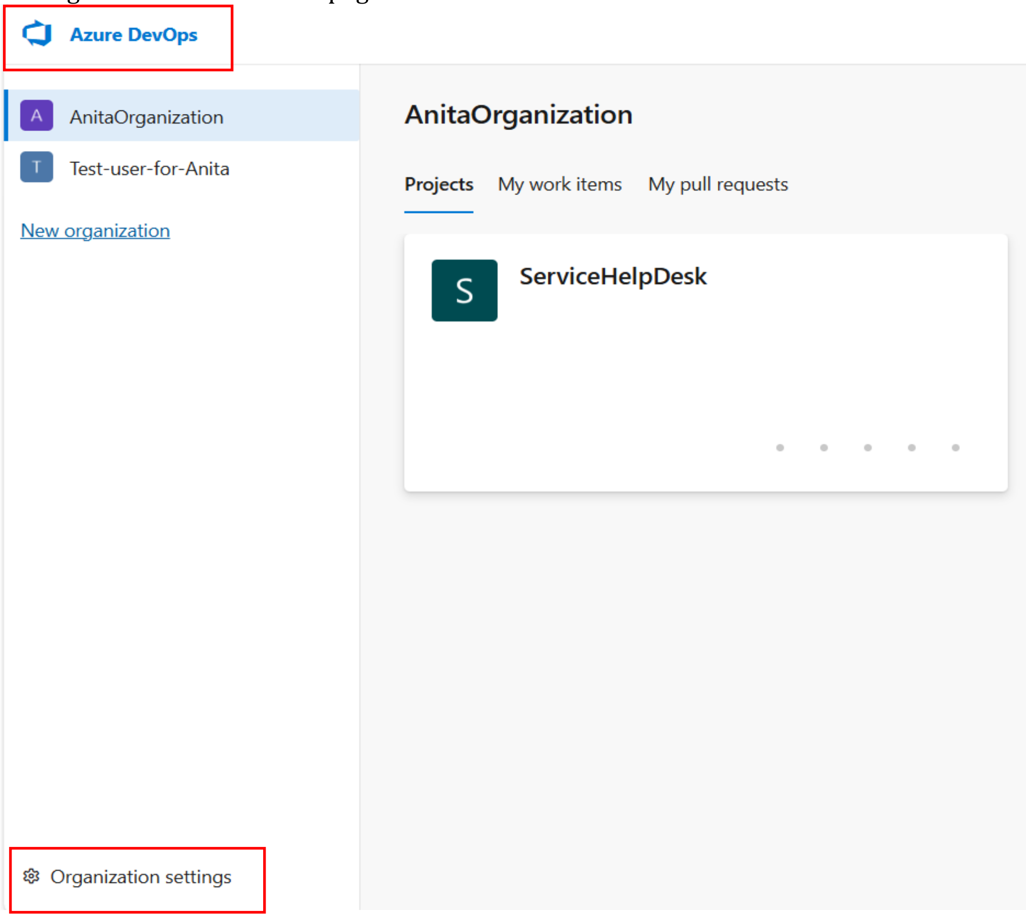


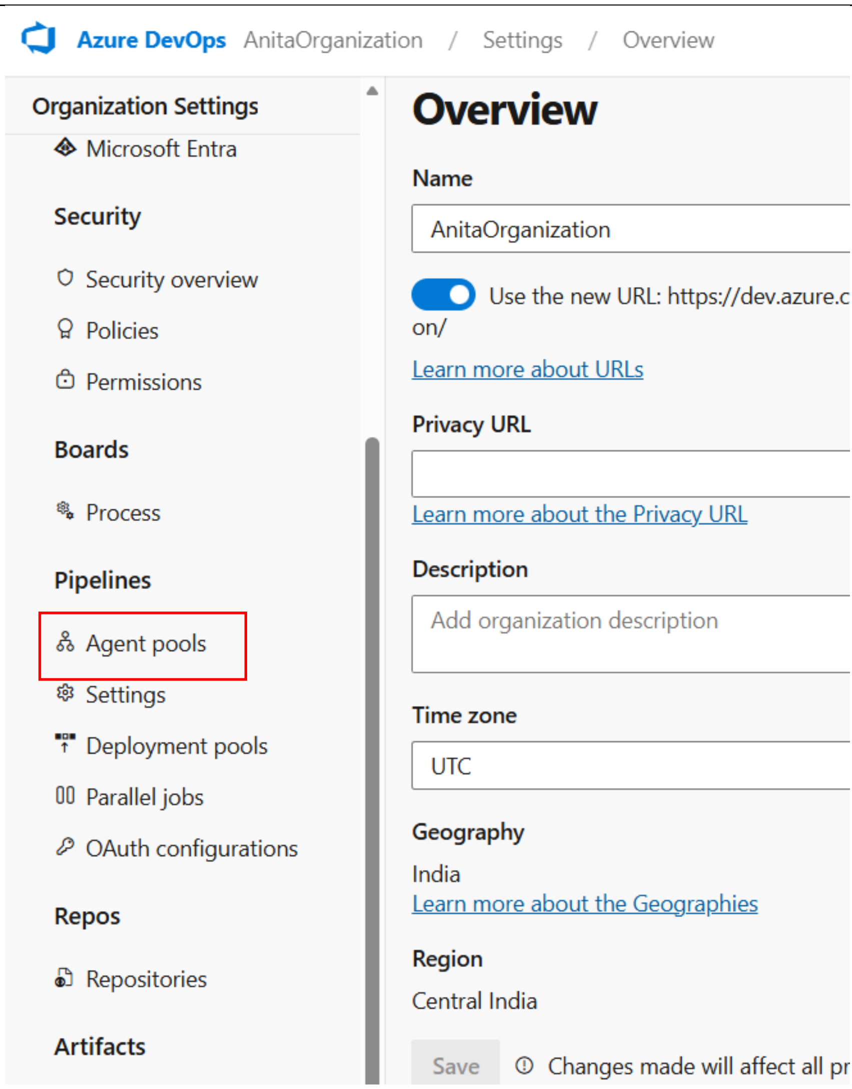
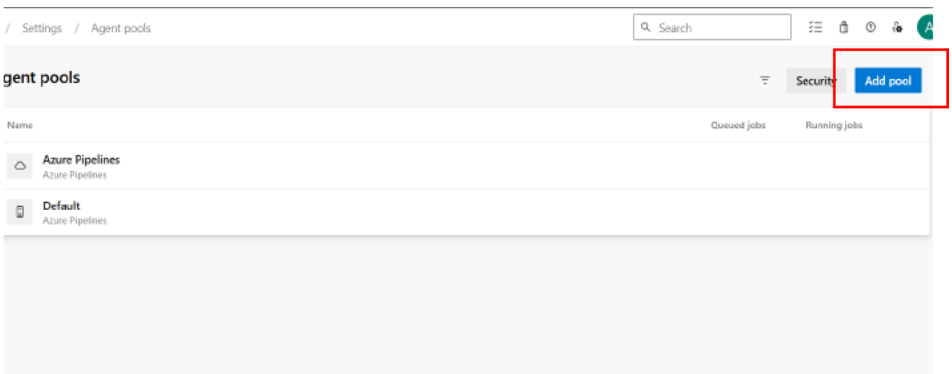
</td>
</tr>
<tr>
<td>4</td>
<td>
Now from left hand menu click on Repos and see the EnvironmentRepo name updated and click on the same.



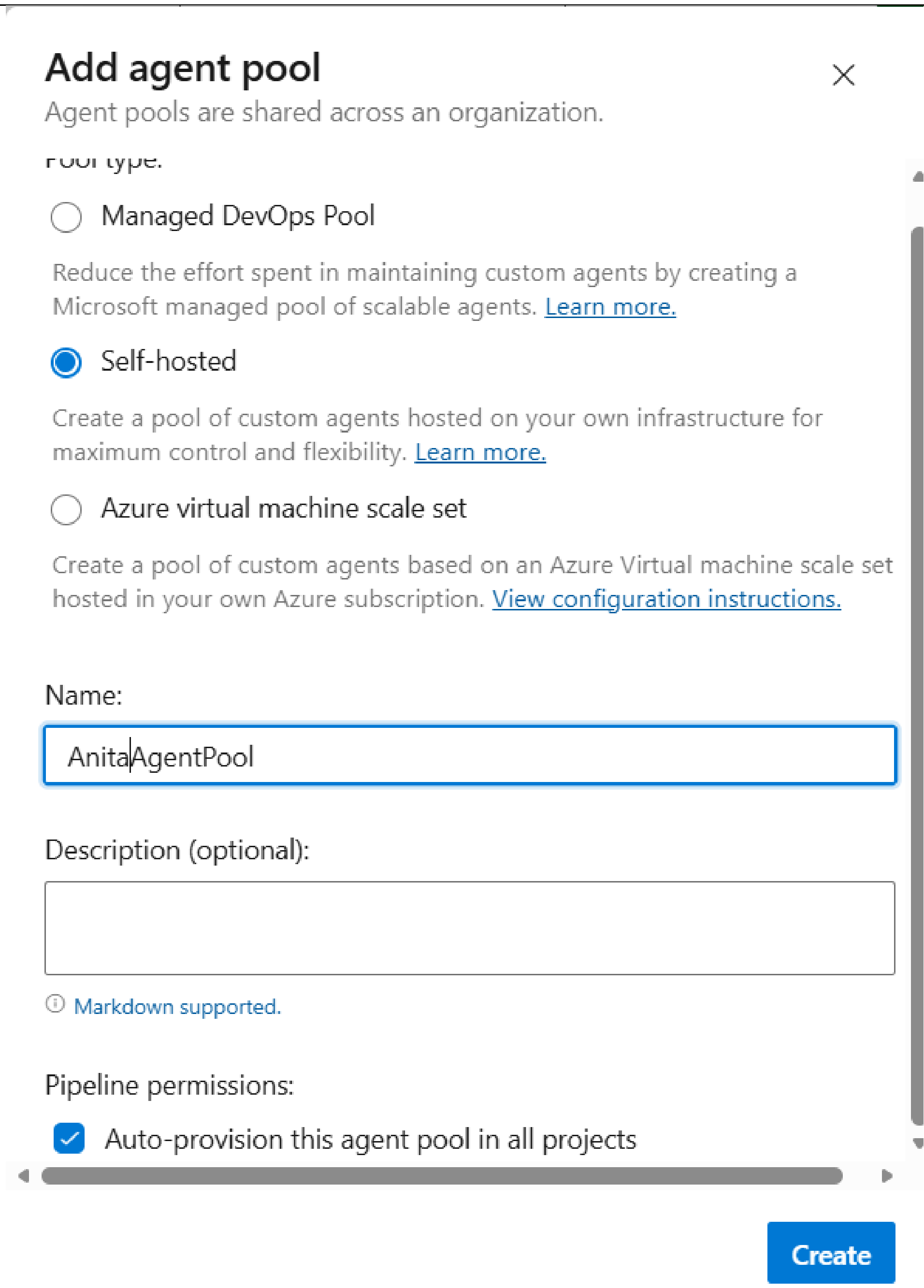
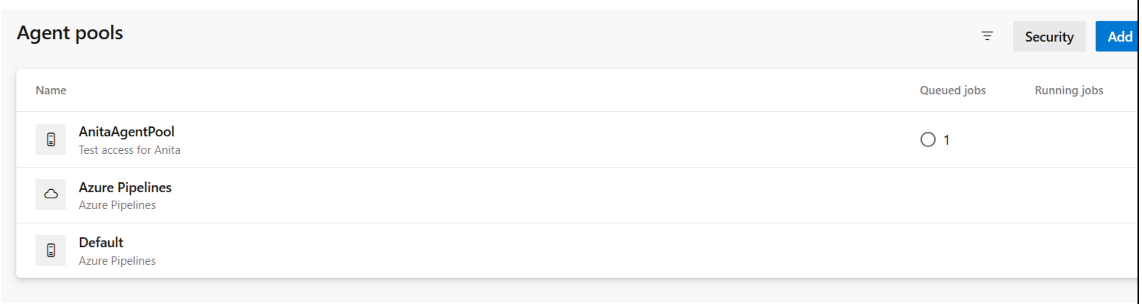
</td>
</tr>
</table>

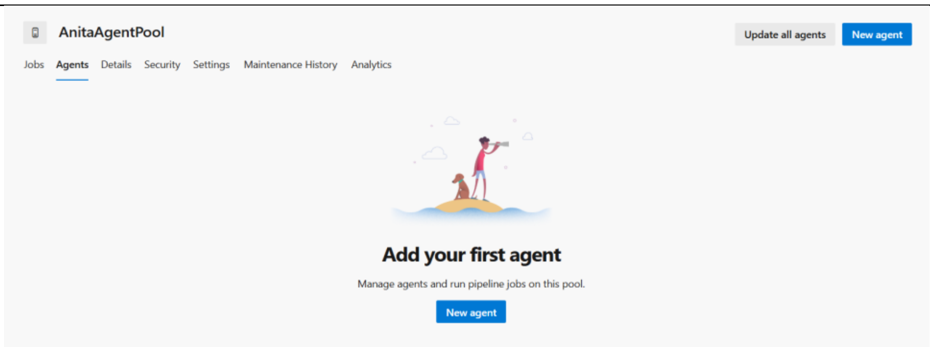
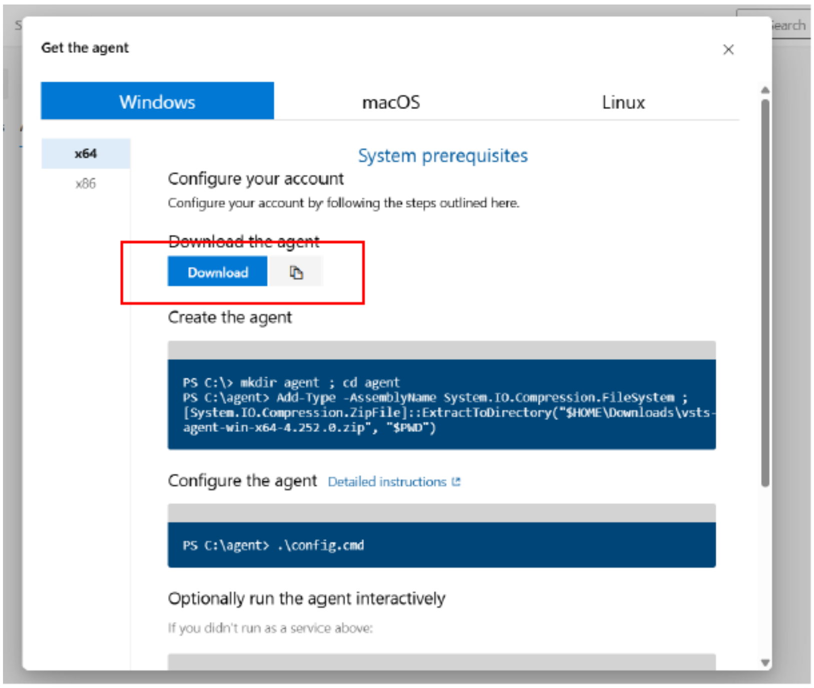
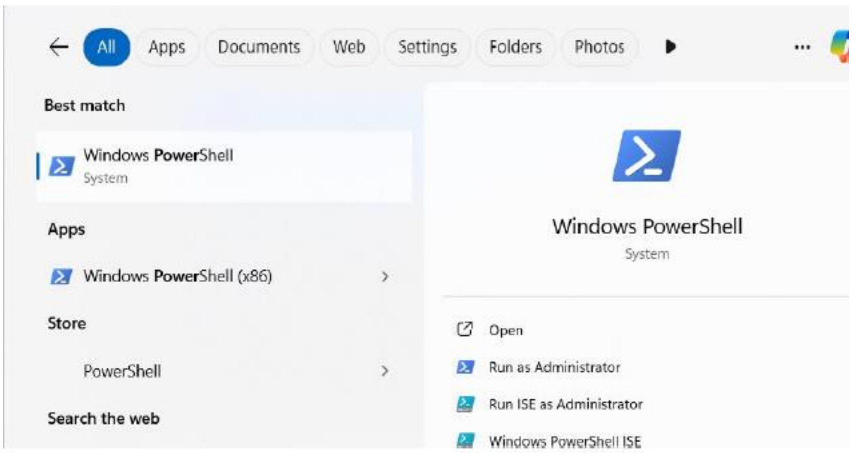
Agent Pool Configuration

1 -	Click on Azure DevOps at the top left, select your organization created and click Organization Settings at the bottom of the page. 
2	Select Agent pool

<table>
<tr><td></td><td>
</td></tr>
<tr><td>3
-</td><td>Click on Add Pool

</td></tr>
<tr><td>4
-</td><td>Provide Required details,
Pool type: Self Hosted

Name : <Yourname>Pool</td></tr>
</table>

	 Click on **Create**
5 -	Select the agent pool created. 
6 -	Select **Agents** and click on **New Agent**

	 Get the Agent, select **Windows,** Click on Download, agent will be downloaded in to Downloads folder in your machine. 
7 -	After downloading, Open **Powershell** in your machine 

Administrator: c:\windows\system32\windowspowershell\v1.0\powershell.exe

```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Users\anita.rananaware>
```

Navigate to C drive; refer commands in below page

cd C:\

```
PS C:\Users\anita.rananaware> cd c:\
PS C:\>
```

Execute the command from the below page,

Create the agent

```
PS C:\> mkdir agent ; cd agent
PS C:\agent> Add-Type -AssemblyName System.IO.Compression.FileSystem ;
[System.IO.Compression.ZipFile]::ExtractToDirectory("$HOME\Downloads\vsts-
agent-win-x64-4.266.2.zip", "$PWD")
```

Configure the agent Detailed instructions

```
PS C:\agent> .\config.cmd
```

Optionally run the agent interactively

If you didn't run as a service above:

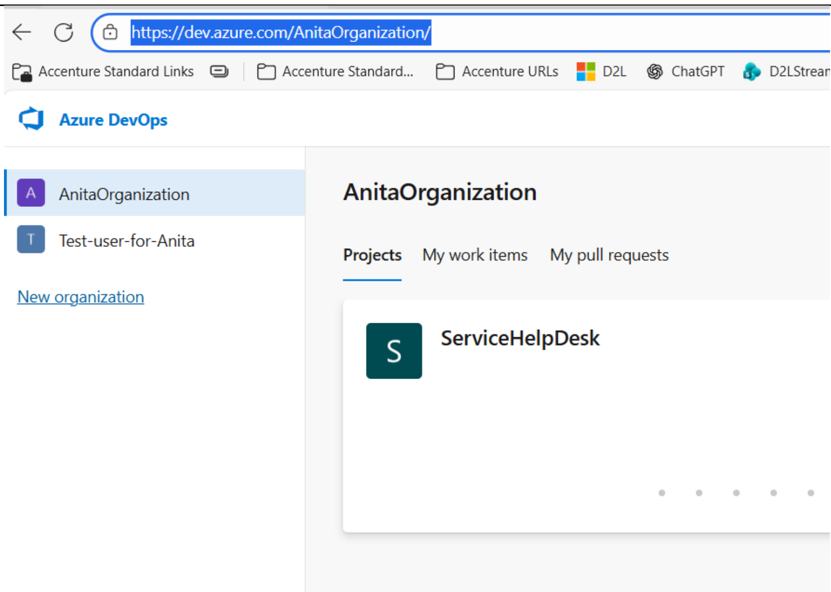
```
PS C:\agent> .\run.cmd
```

That's it!

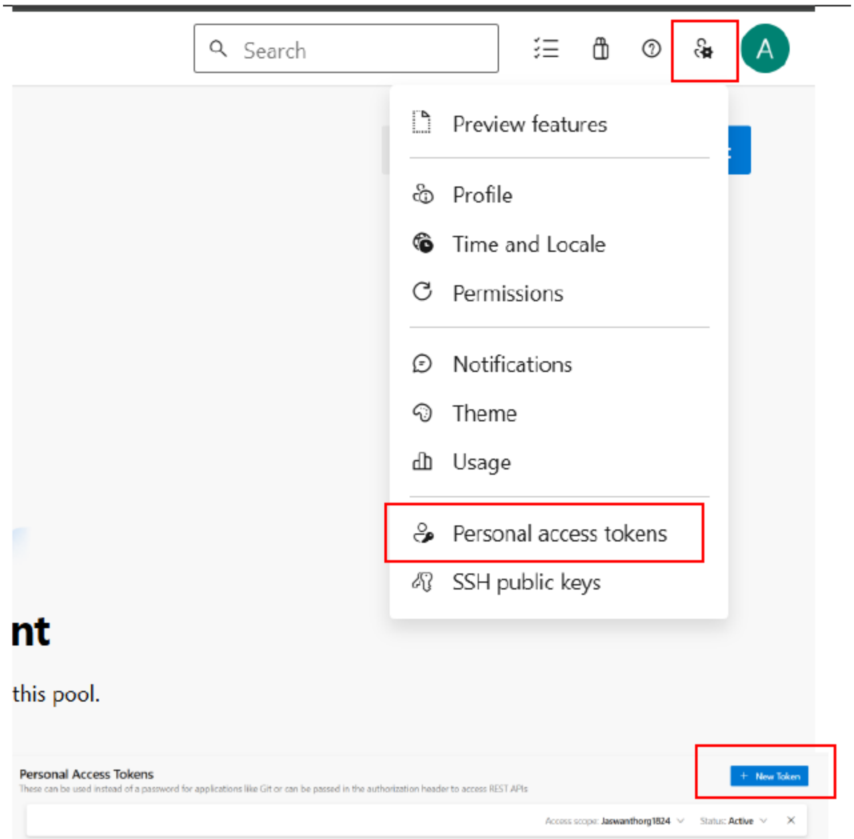
More Information

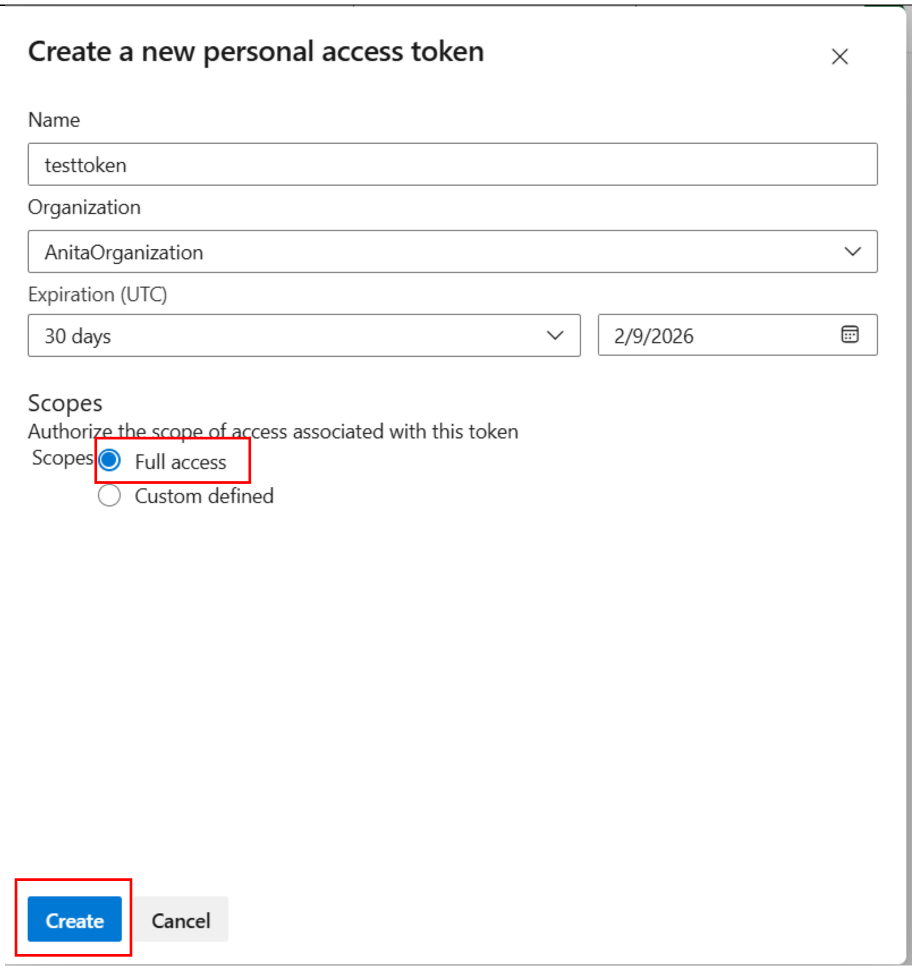
```
PS C:\> mkdir agent ; cd agent

    Directory: C:\

Mode                 LastWriteTime         Length Name
----                 -------------         ------ ----
d-----         3/14/2025  12:09 PM                agent

PS C:\agent> Add-Type -AssemblyName System.IO.Compression.FileSystem ; [System.IO.Compression.ZipFile]::ExtractToDirectory("$HOME\Downloads\vsts-agent-win-x64-4.252.0.zip", "$PWD")
PS C:\agent>
```

Run .\config.cmd
Provide below details
Enter server URL=> **Ex URL**: https://dev.azure.com/AnitaOrganization/
Ref ss:



Enter authentication type (press enter for PAT) => press **Enter**
Enter personal access token => **<Give pat token here>**
To generate pat token follow below steps:





Copy the pat token and save it somewhere for later use.
Enter agent pool (press enter for default) > **<give pool name> Ex: AnitaAgentPool**

Enter work folder (press enter for _work) => press **Enter**
Enter run agent as service? (Y/N) (press enter for N) => Press **Enter**
Enter configure autologon and run agent on startup? (Y/N) (press enter for N) => Press **Enter**

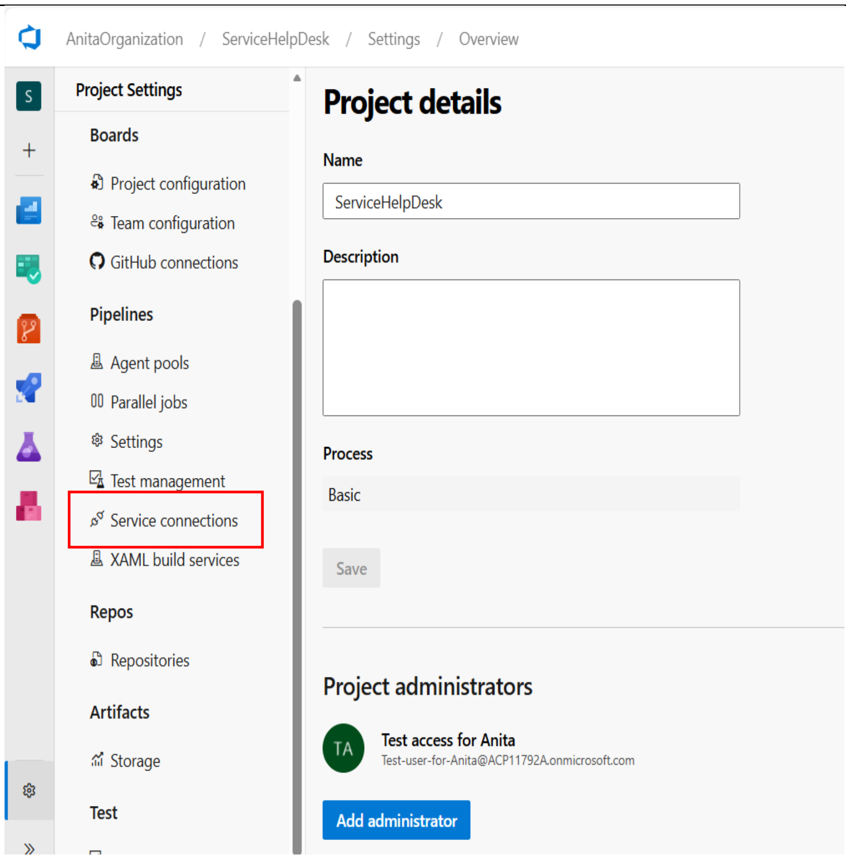
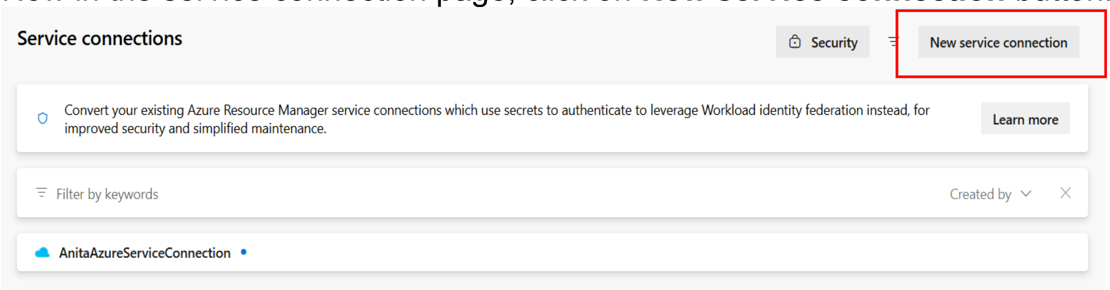
```
C:\agent>.\config.cmd

  ___                      ______ _            _ _
 / _ \                     | ___ (_)          | (_)
/ /_\ \_____   _ _ __ ___  | |_/ /_ _ __   ___| |_ _ __   ___  ___
|  _  |_  / | | | '__/ _ \ |  __/| | '_ \ / _ \ | | '_ \ / _ \/ __|
| | | |/ /| |_| | | |  __/ | |   | | |_) |  __/ | | | | |  __/\__ \
\_| |_/___|\__,_|_|  \___| \_|   |_| .__/ \___|_|_|_| |_|\___||___/
                                   | |
        agent v4.264.2             |_|          (commit 491eca9)


>> Connect:

Enter server URL > https://dev.azure.com/AnitaOrganization/
Enter authentication type (press enter for PAT) >
Enter personal access token > ************************************************************
Connecting to server ...

>> Register Agent:

Enter agent pool (press enter for default) > AnitaAgentPool
Enter agent name (press enter for PDC2C-L-4VL18S3) > AnitaAgent
Scanning for tool capabilities.
Connecting to the server.
Successfully added the agent
Testing agent connection.
Enter work folder (press enter for _work) > Anita
2026-01-09 18:51:42Z: Settings Saved.
Enter run agent as service? (Y/N) (press enter for N) >
Enter configure autologon and run agent on startup? (Y/N) (press enter for N) >

C:\agent>
```

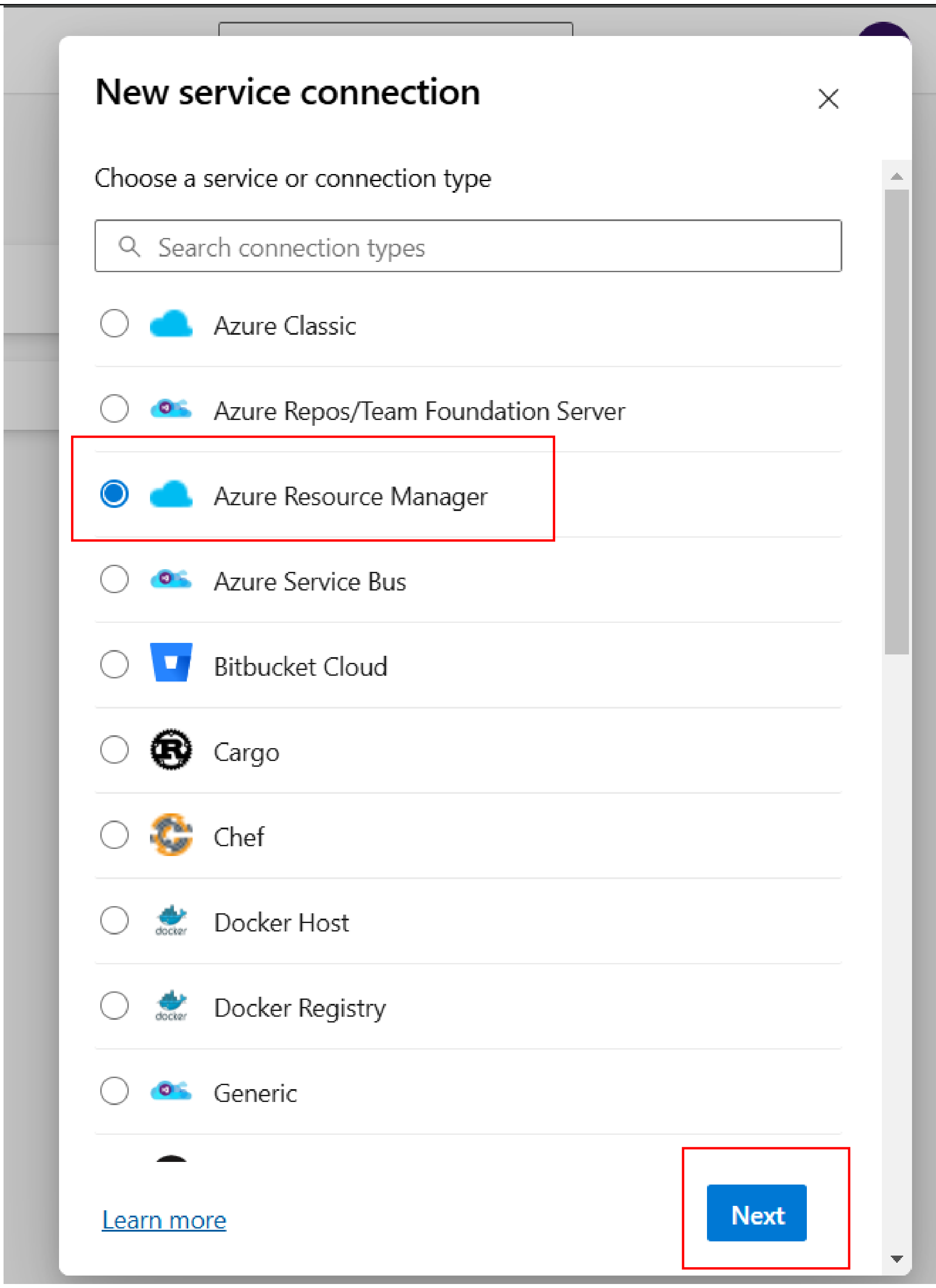
Execute below command **.\run.cmd**

```
C:\agent>.\run.cmd
Scanning for tool capabilities.
Connecting to the server.
2026-01-09 18:53:29Z: Listening for Jobs
```

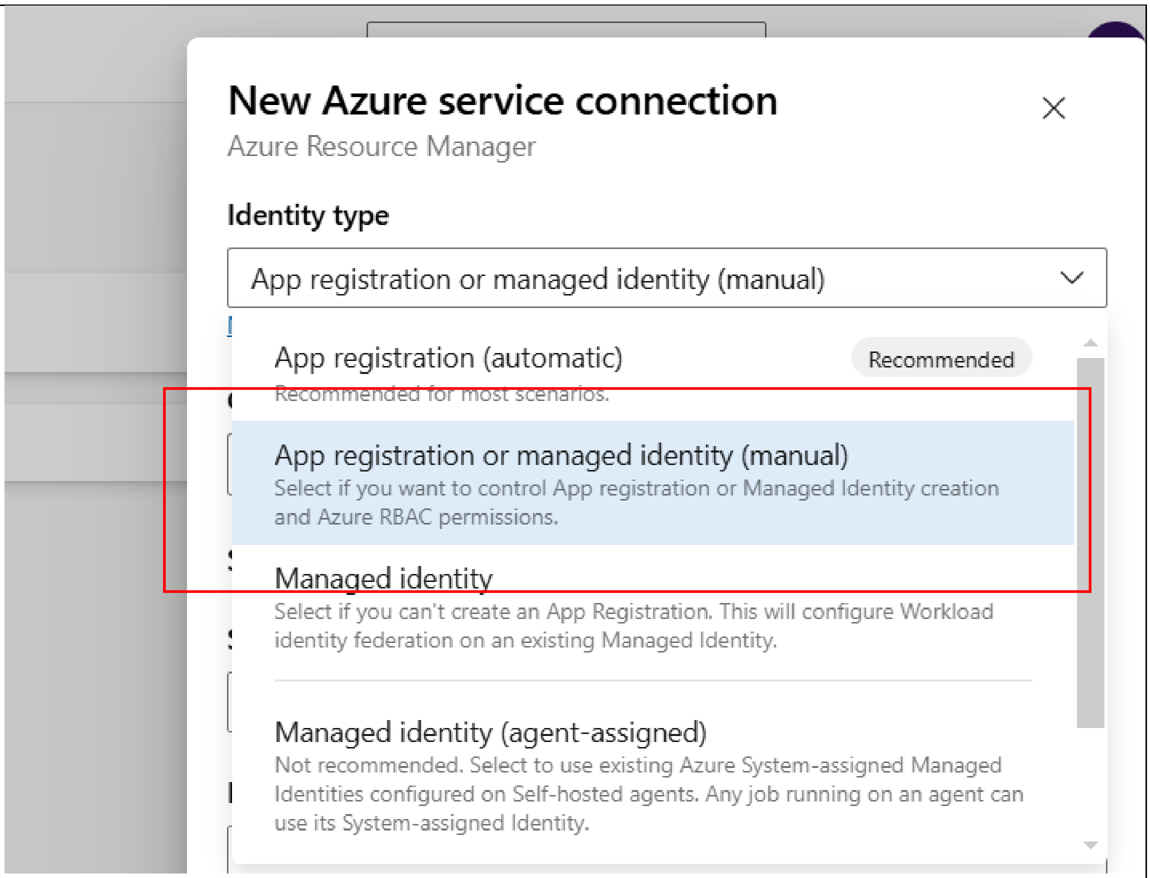
Service connection set up for Azure Service

1	Go to Project settings, from the left menu, Under Pipelines, click on Service Connections

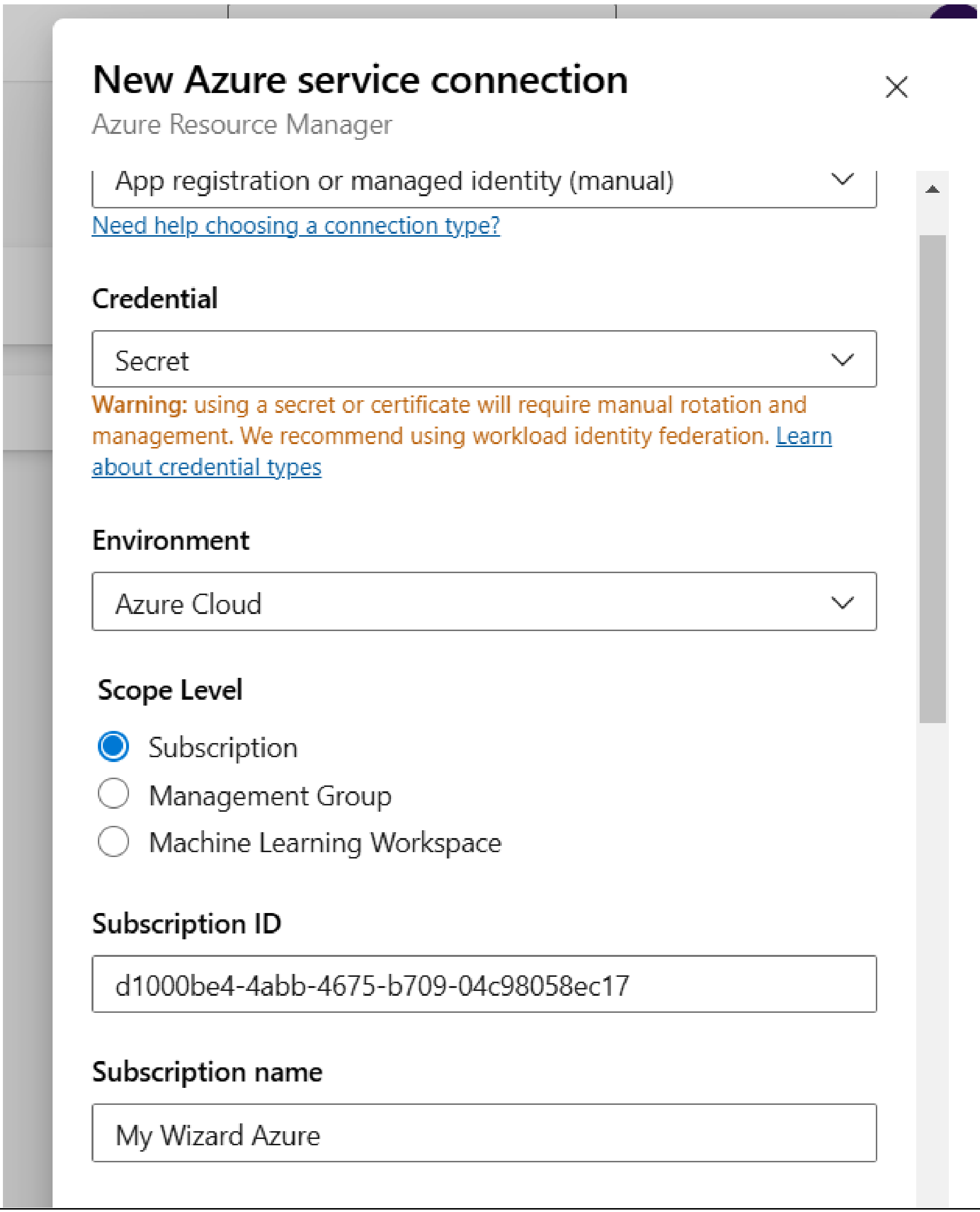
<table>
<tr><td></td><td>
</td></tr>
<tr><td>2</td><td>Now in the service connection page, click on New service connection button.

</td></tr>
<tr><td>3</td><td>In New service connection page, select Azure Resource Manager option and click on Next</td></tr>
</table>

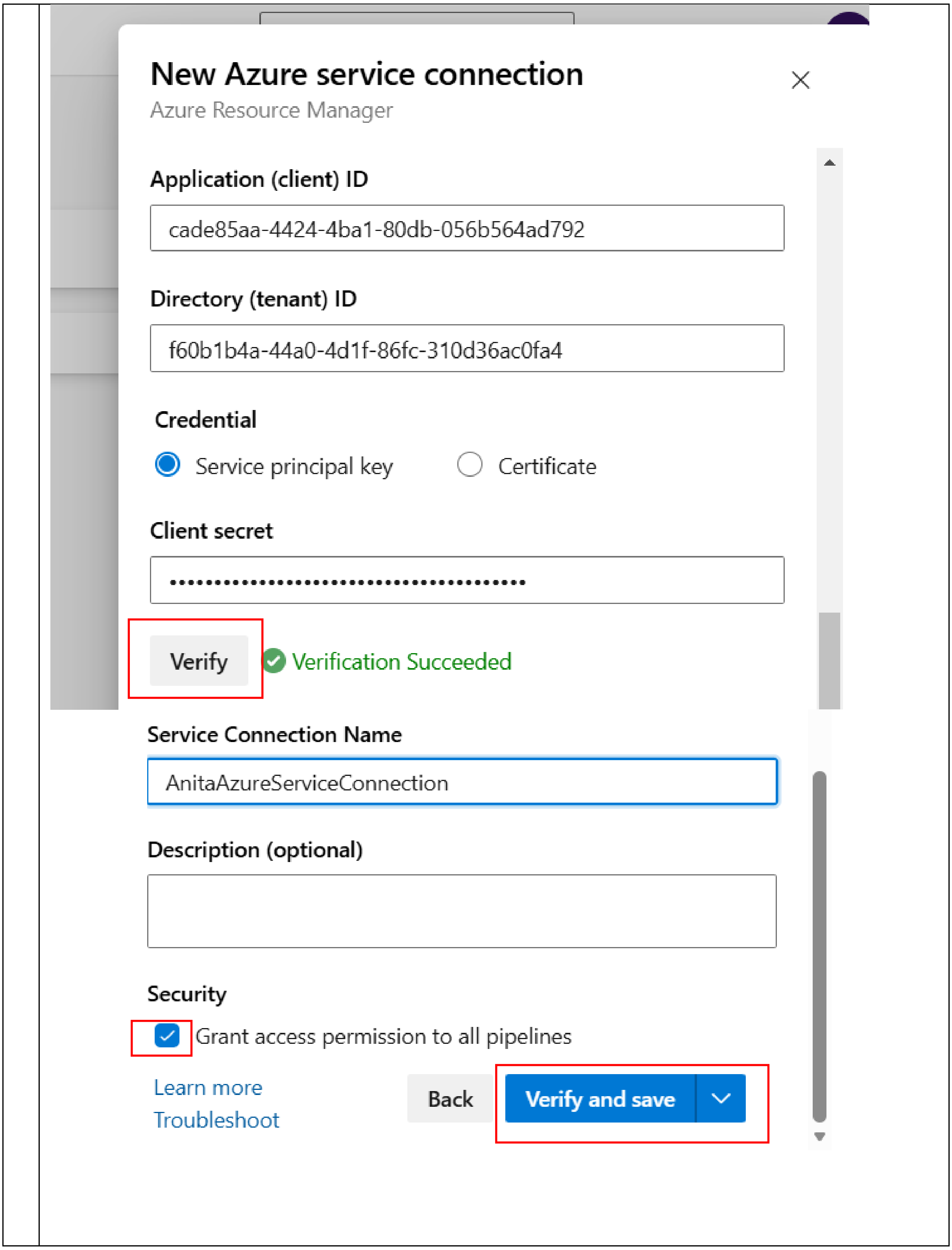


3	Select Identity type **App Registration or managed identity (manual)** option

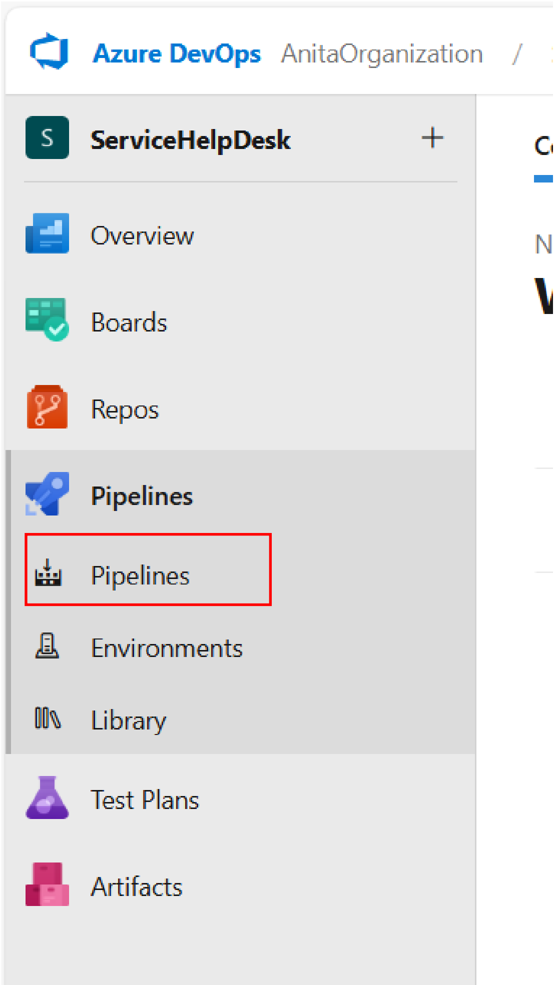
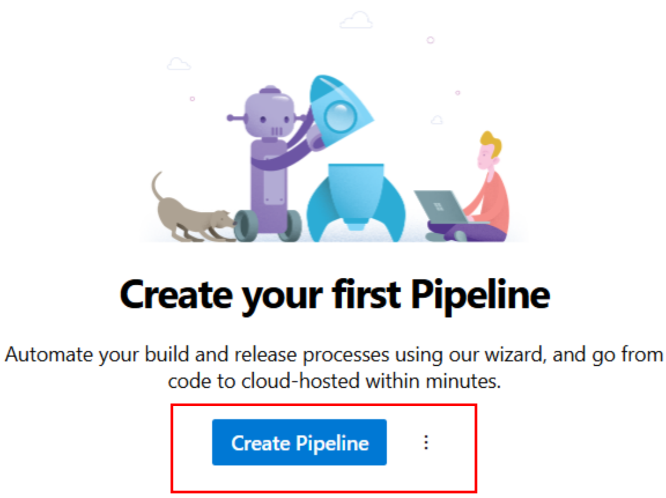
	
4	Enter / Select the options as given. Highlighted values are provided to you by coherts team already. Credential : as Secret (from drop down) Environment : Azure Cloud Scope level : Subscription Subscription Id : <Enter the provided value> Subscription name : <Enter the provided value> Application (client) ID: <Enter the provided value> Directory (tenant) ID: <Enter the provided value> Credential: select Service principal key Client secret: <Enter the provided value> Provide service connection name as <yourname>serviceconnection as shown below After entering **Client secret** click on Verify and ensure Verification succeeded. Give the service connection name as <yourname>AzureServiceConnection. For example AnitaAzureServiceConnection and same can be given in description. Ensure check box is checked under security and click on **verify and save** button.

5	Ensure there are two service connections under your project

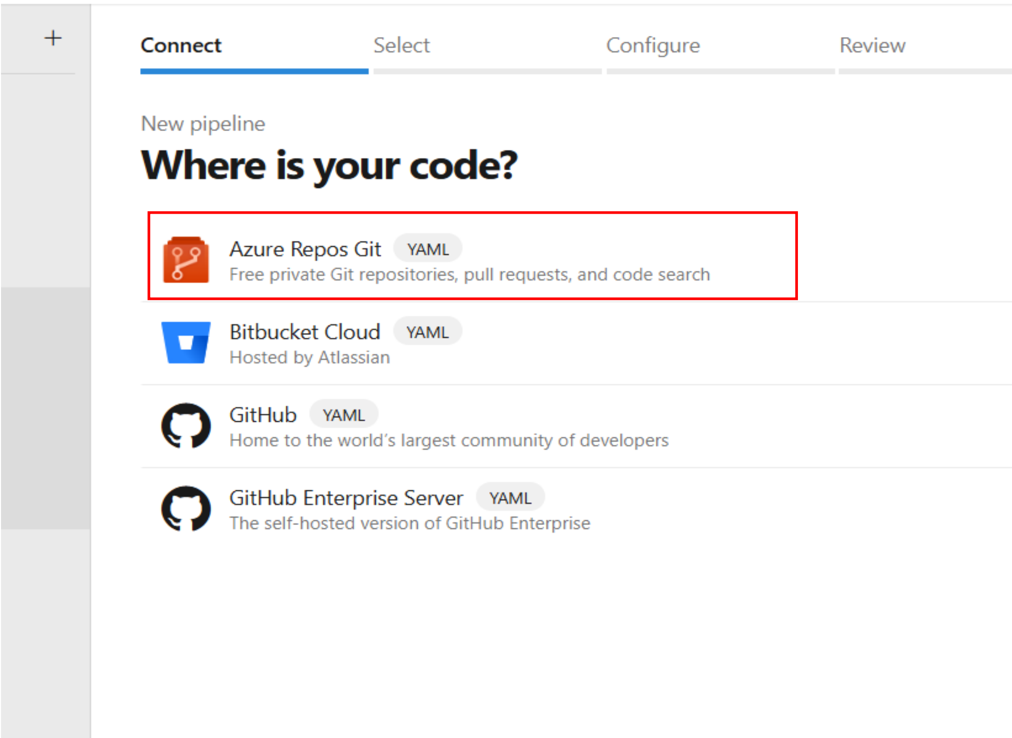



New Azure service connection
Azure Resource Manager
Application (client) ID
cade85aa-4424-4ba1-80db-056b564ad792
Directory (tenant) ID
f60b1b4a-44a0-4d1f-86fc-310d36ac0fa4
Credential
Service principal key
Certificate
Client secret
Verify
Verification Succeeded
Service Connection Name
AnitaAzureServiceConnection
Description (optional)
Security
Grant access permission to all pipelines
Learn more
Troubleshoot
Back
Verify and save

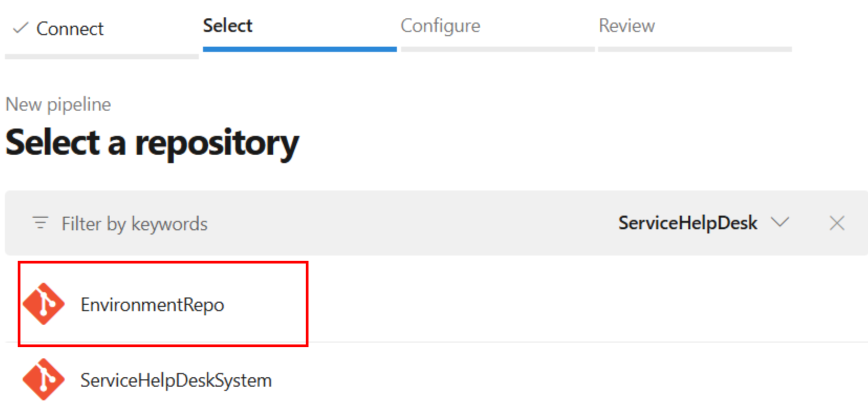
Variable Setup and Environment Pipeline

1	From the left menu option, click on Pipeline icon and select **Pipelines** option. 
2	Click on **create pipeline** button 

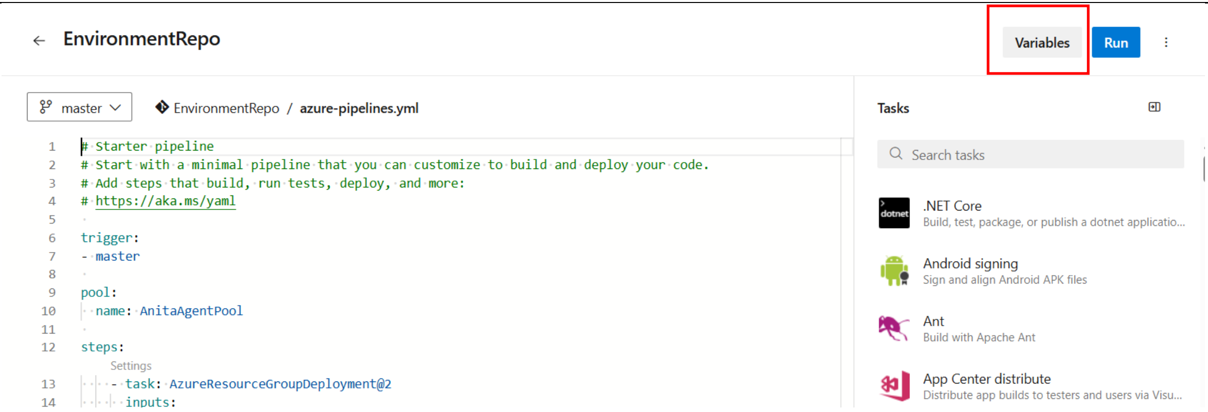
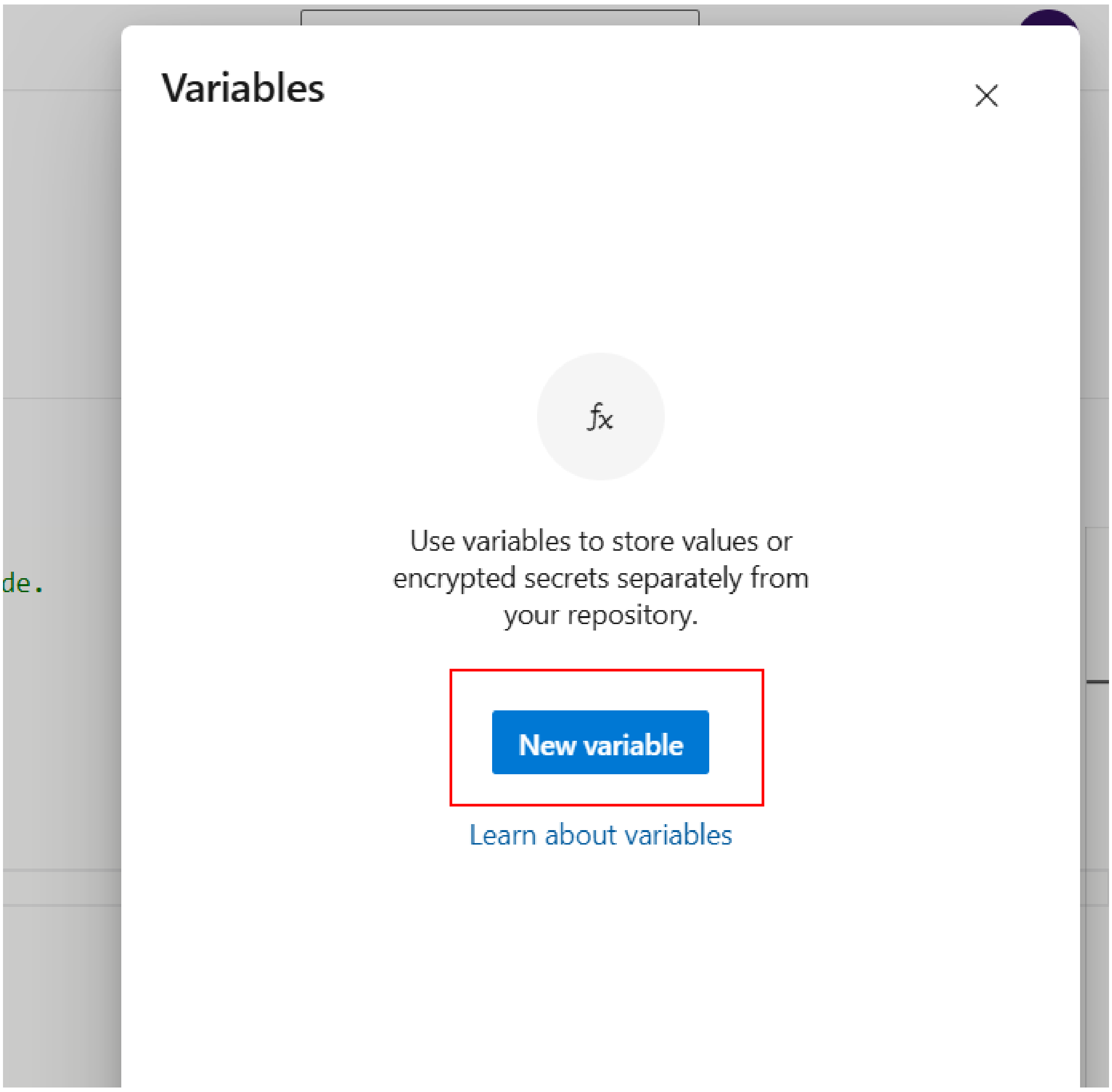
<table>
<tr><td></td><td></td></tr>
<tr><td>3</td><td>In Where is your code? Section, select the option Azure Repos Git



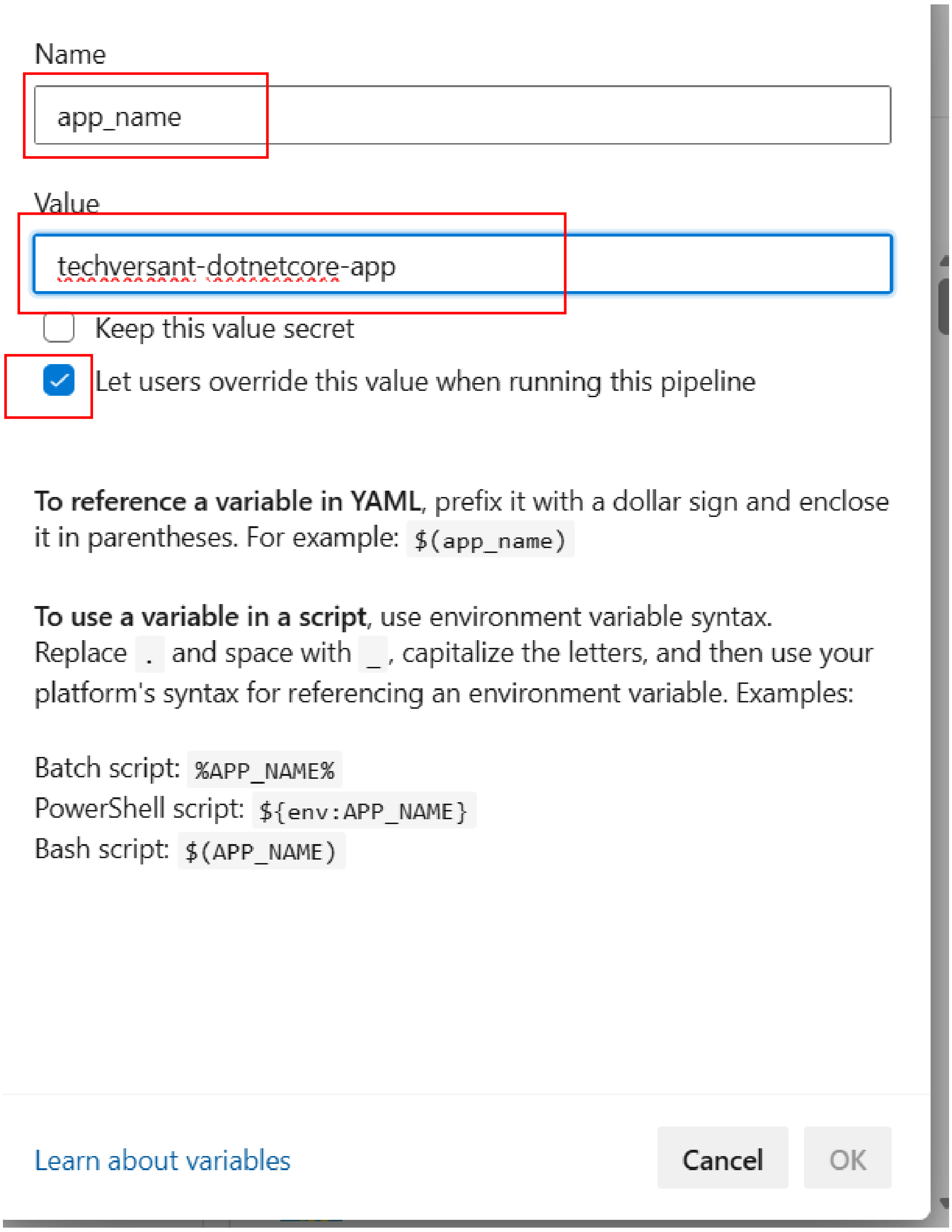
</td></tr>
<tr><td>4</td><td>In next page, select the repository name EnvironmentRepo



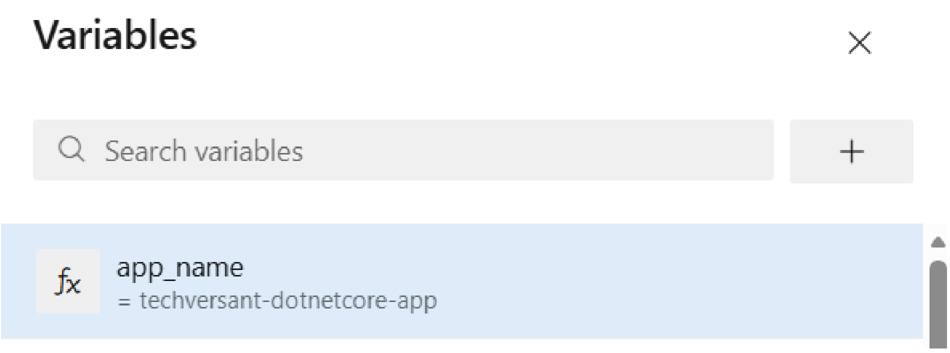
</td></tr>
<tr><td>5</td><td>Now click on Variables box</td></tr>
</table>

<table>
<tr><td></td><td>
</td></tr>
<tr><td>6</td><td>Click New Variable button

</td></tr>
<tr><td>7</td><td>In the new variable screen, enter as follows

Name : app_name
Value : techversant-dotnetcore-app
Check the checkbox as shown below. And finally click on ok button.</td></tr>
</table>



Observe the variable added as shown below

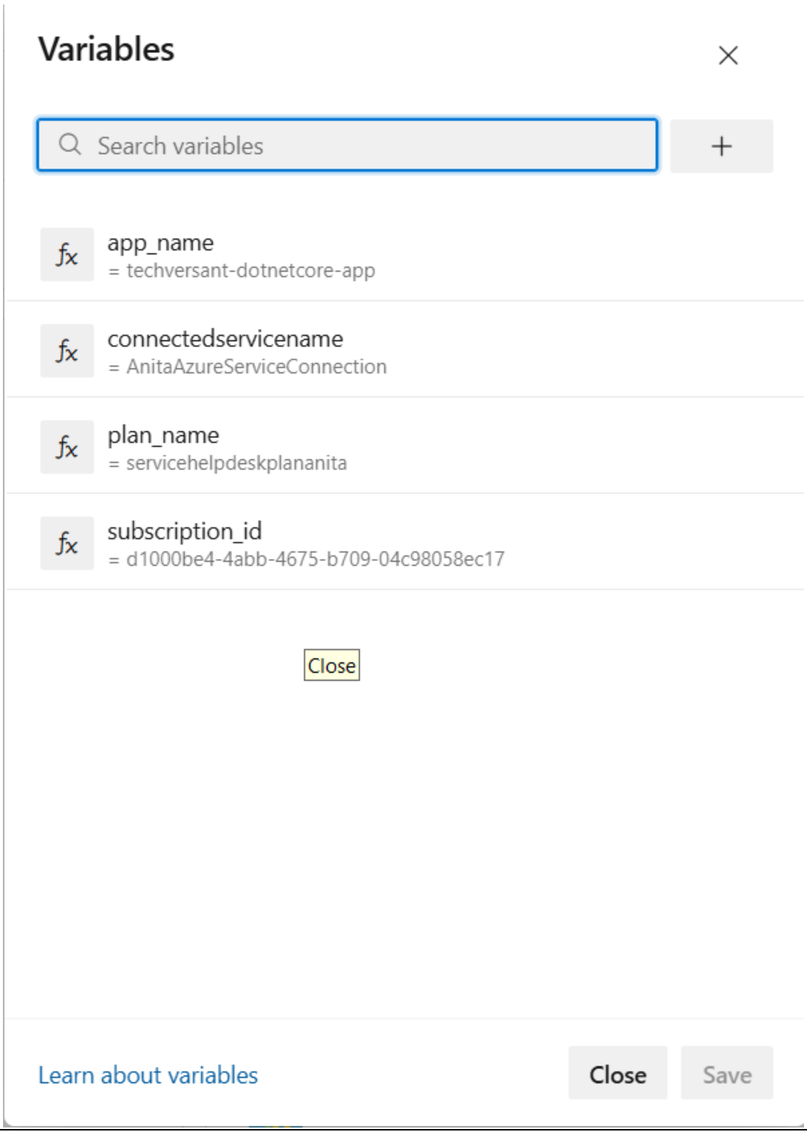


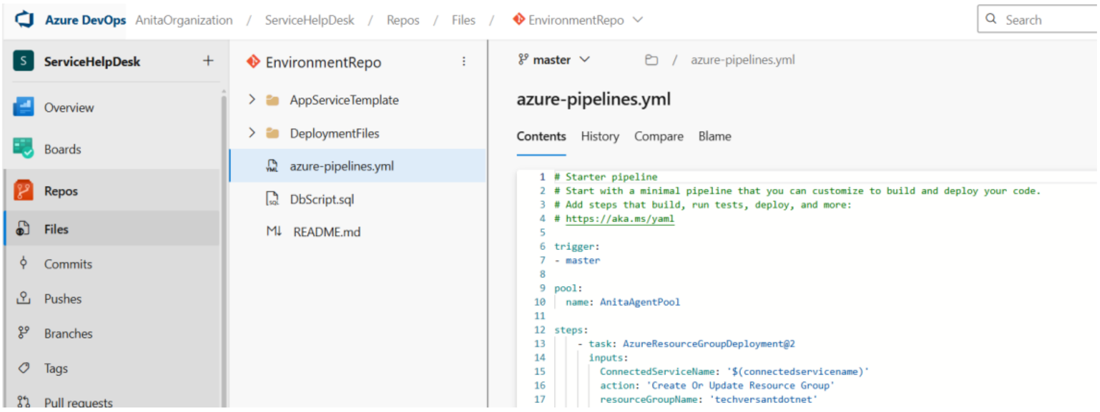
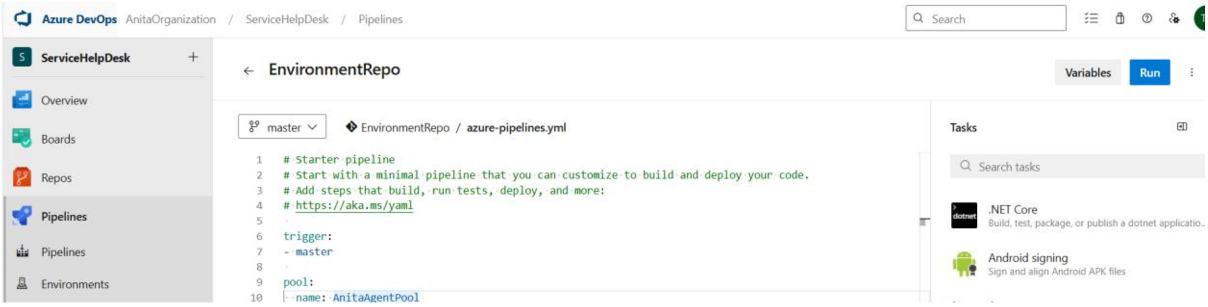
8 Now click on + symbol to add variables one by one as per below table. Wherever <name> is mentioned use your name.

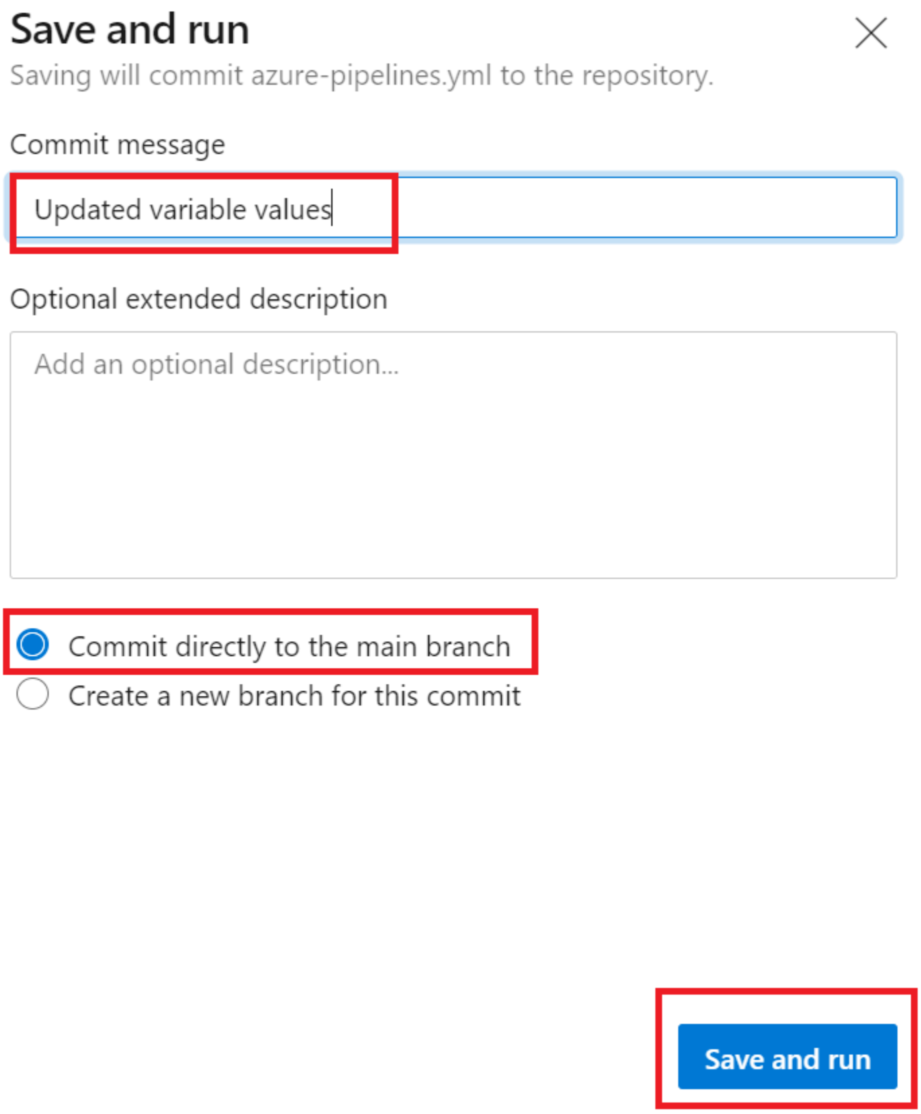
First row variable addition is already done in step number 7

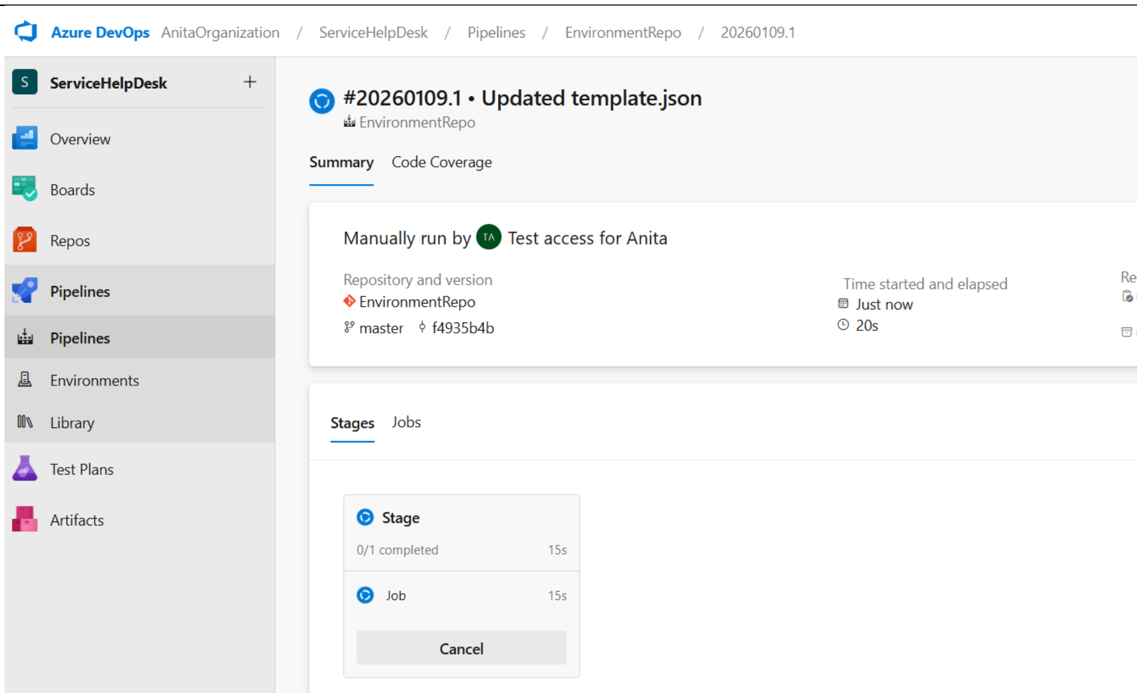
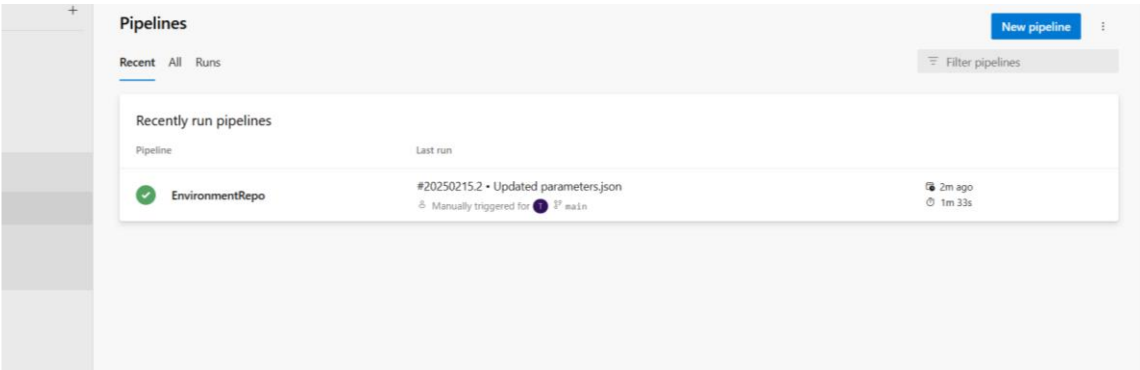
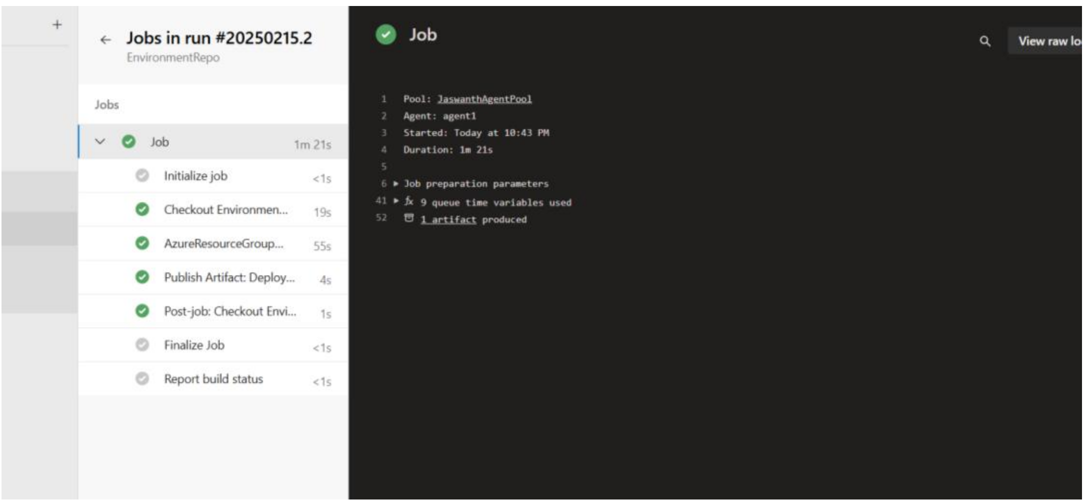
Name	Value
app_name	techversant-dotnetcore-app
connectedservicename	Refer Appendix to get the name. Example : **AnitaAzureServiceconnection**
plan_name	**techversant-plan**
subscription_id	<Provided by ops team> Example : b4xxxxxx-exx2-4xxxd-axx-6xxxxxxxxxxx Not giving the full ID here due to security policy.

9 After entering all the variables click on save button



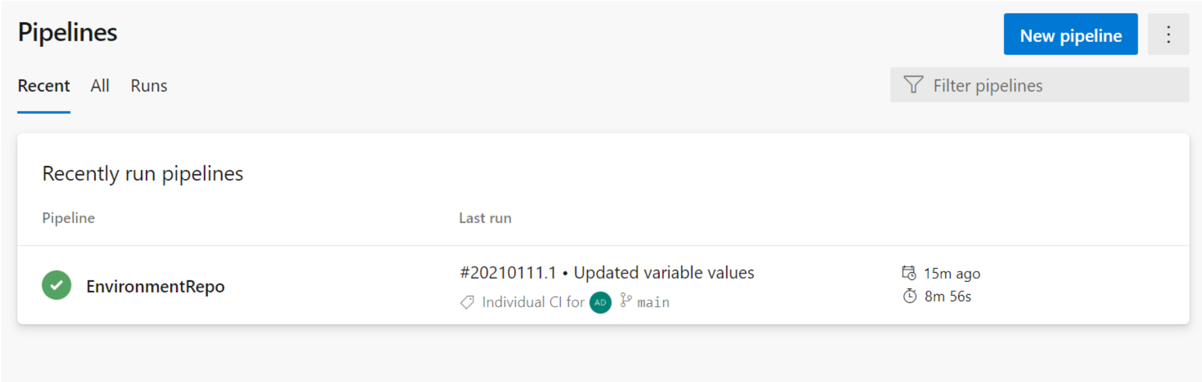
<table>
<tr><td>10</td><td>Open the file azure-pipelines.yml and click Edit button. Replace the AgentPoolName with the agent pool name which you created and click on Commit button.

</td></tr>
<tr><td>11</td><td>After saving the variable values successfully click on save and run button

</td></tr>
<tr><td>12</td><td>In Save and run window enter any message under commit message box. Here it is given updated variable values. Ensure the radio button commit to the main branch is checked and Click on Save and Run</td></tr>
</table>

	
12	Observe the below screen

<table>
<tr><td></td><td>
</td></tr>
<tr><td>13</td><td>Click on Job link after the blue color turn to green icon

</td></tr>
<tr><td>15</td><td>This execution takes 10-15 minutes of time. It varies depends on your network bandwidth. The pipeline should be executed exactly once.
Ensure all the jobs are executed successfully.

</td></tr>
</table>

<table>
<tr><td></td><td>
If you wish to see the resources deployed, click on show deployed resource under the jobs which turned green and view the resource in the cmd window on the right.

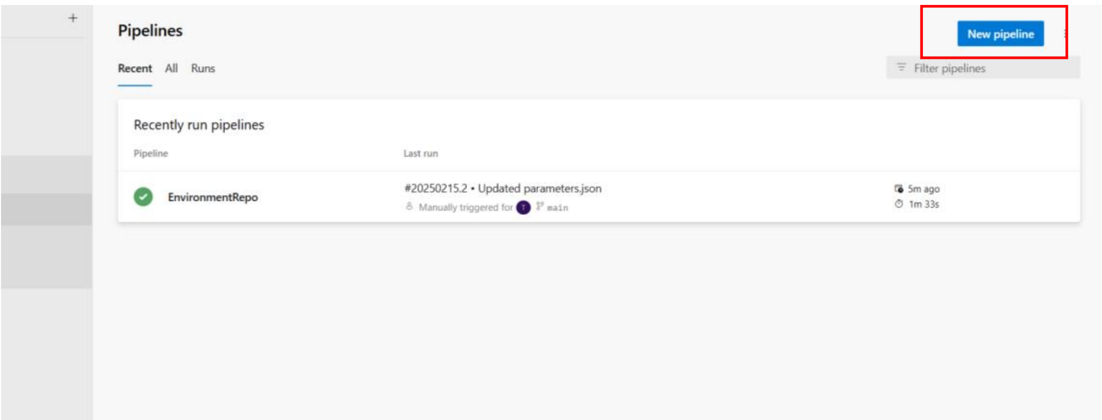
Click on Pipelines from the left menu to view the EnvironmentRepo pipeline



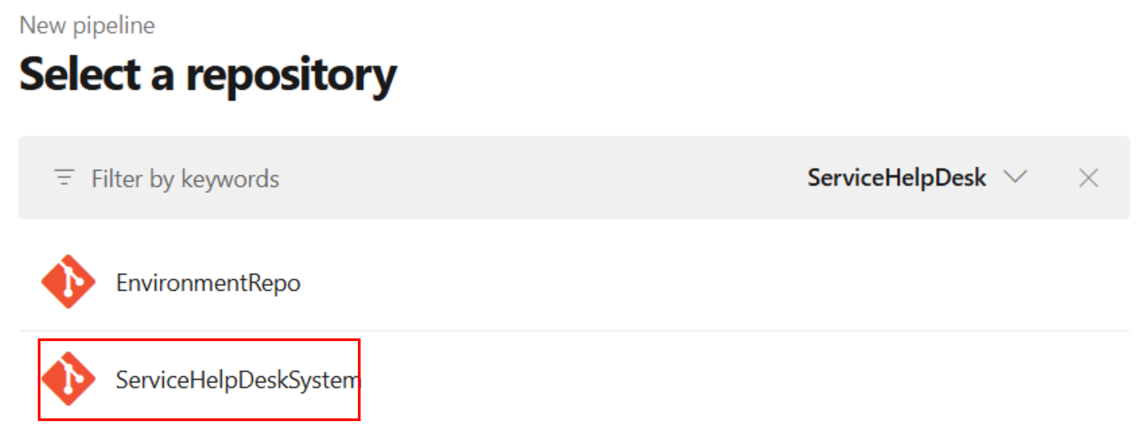
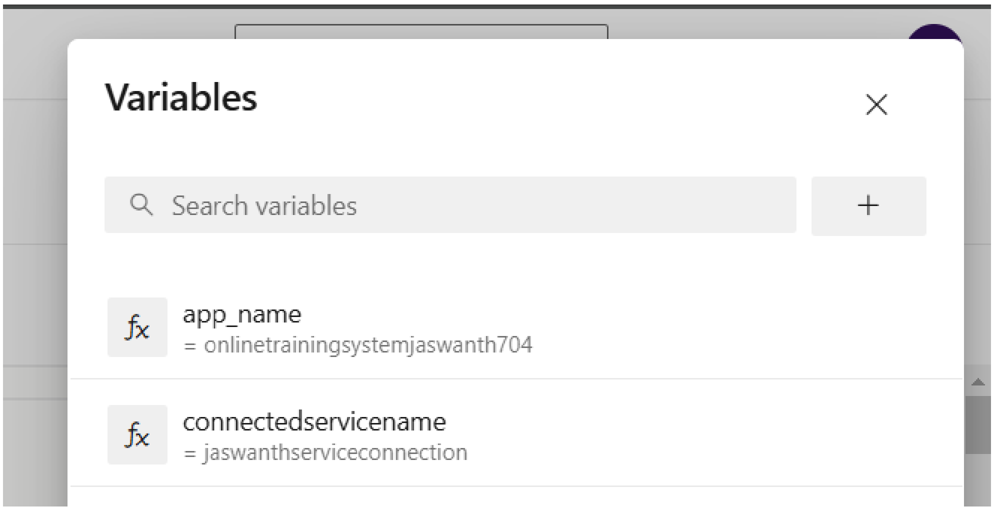
</td></tr>
</table>

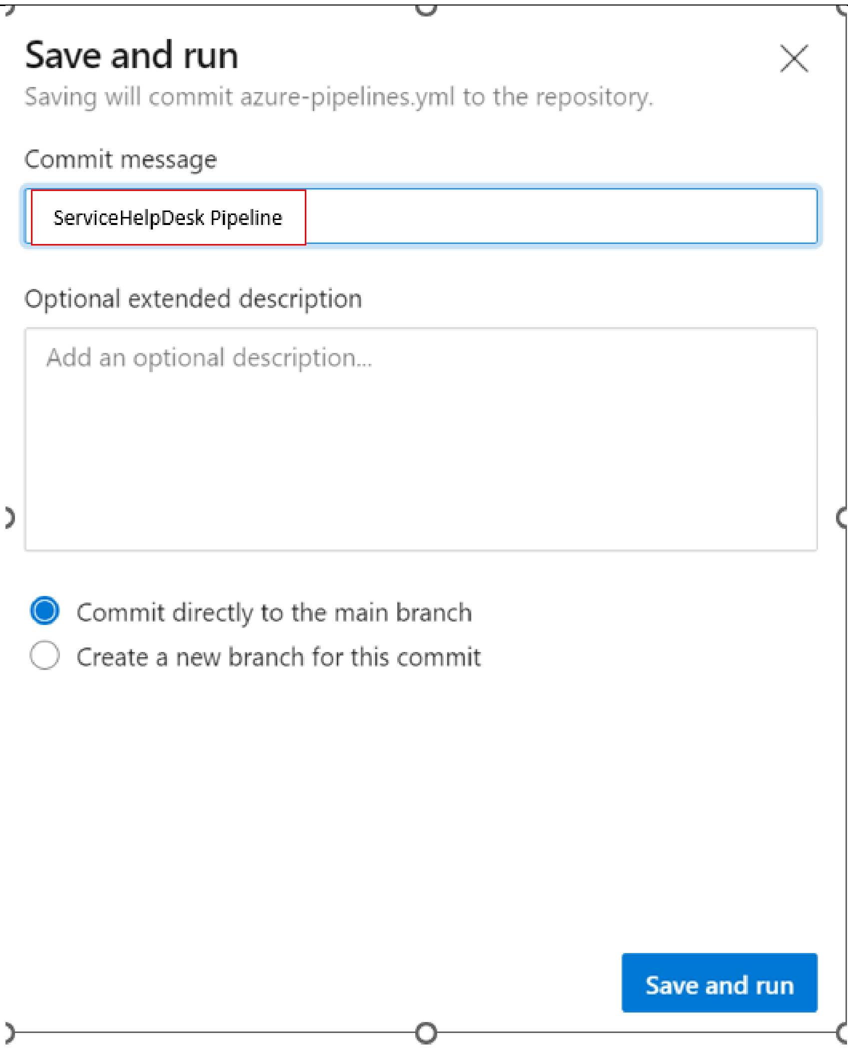
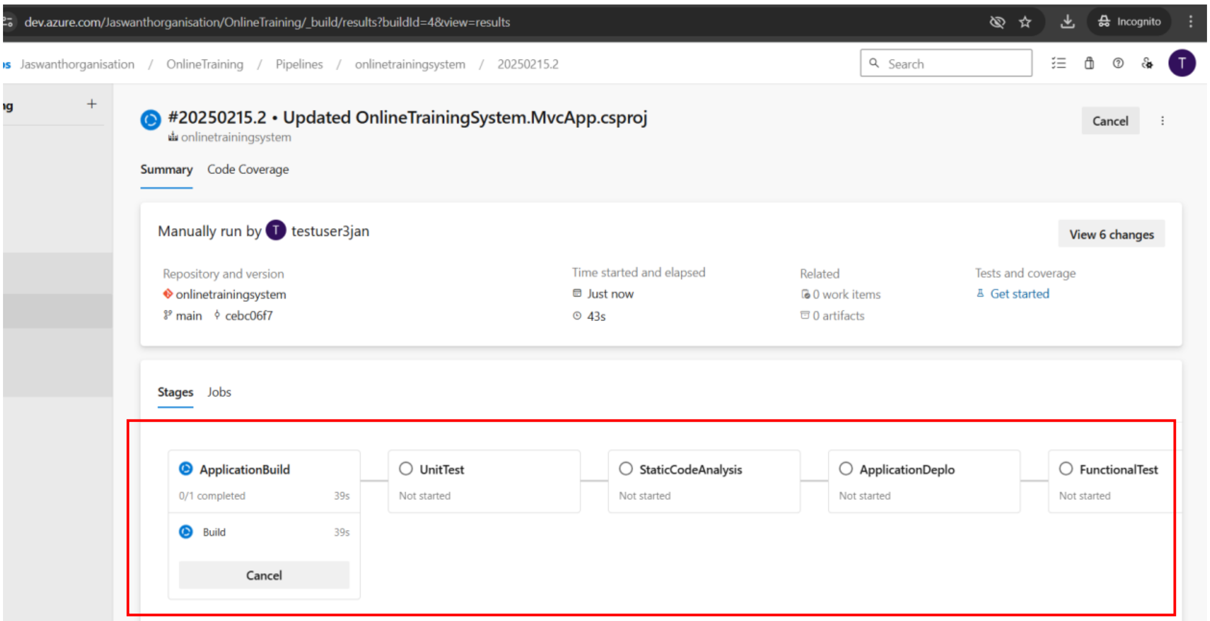
ServiceHelpDeskSystem Pipeline Execution

<table>
<tr><td>1</td><td>
Now click on New Pipeline button to create one more pipeline for Deploying web application

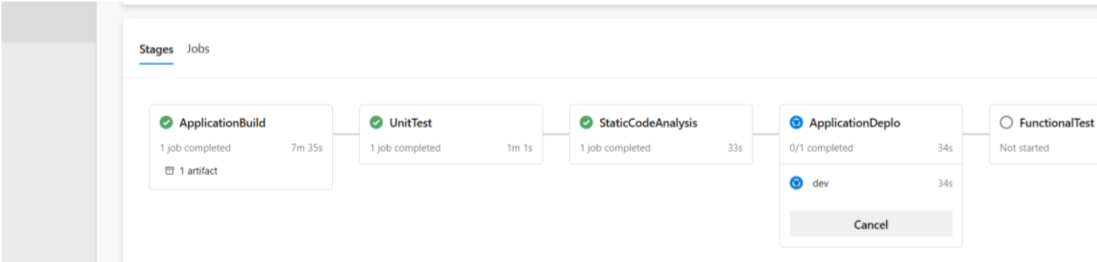


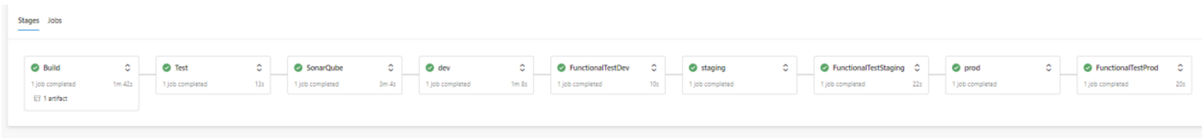
</td></tr>
<tr><td>2</td><td>
In Where is your code? Section select Azure Repos Git
In Select Repository section select ServiceHelpDeskSystem
</td></tr>
</table>

<table>
<tr><td></td><td>
</td></tr>
<tr><td>3</td><td>In the displayed window, click on Variables button and add the variables similar to the previous pipeline step.
<table>
<tr><th>Name</th><th>Value</th></tr>
<tr><td>app_name</td><td>App name given in the previous section.
Example : techversant-dotnetcore-app</td></tr>
<tr><td>connectedservicename</td><td>Connectedservicename given in the previous section
Example : AnitaAzureserviceconnection</td></tr>
</table></td></tr>
<tr><td>4</td><td>After adding all the variables click on Save button

</td></tr>
<tr><td>5</td><td>Open the file azure-pipelines.yml and click Edit button. Replace the AgentPoolName with the agent pool name which you created and click on Commit button</td></tr>
<tr><td>5</td><td>Click on save and run button and enter some commit message and click save and run to trigger the pipeline</td></tr>
</table>

	
6	Observe the pipeline stage execution started as shown below 

<table>
<tr><td>7</td><td>Observe the pipeline execution status and wait for the entire pipeline stages to complete (turn green)

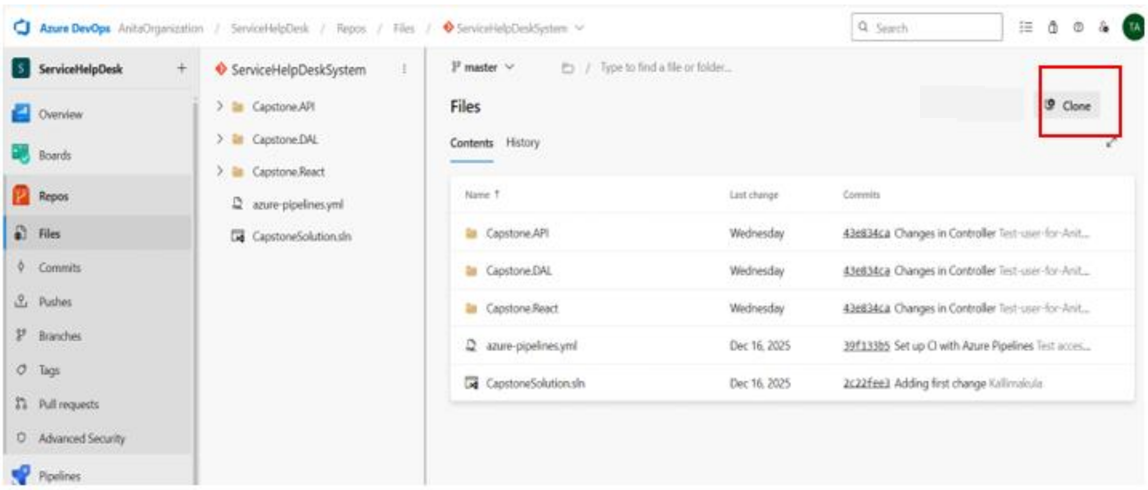

</td></tr>
<tr><td>11</td><td>Complete pipeline is executed and shown below

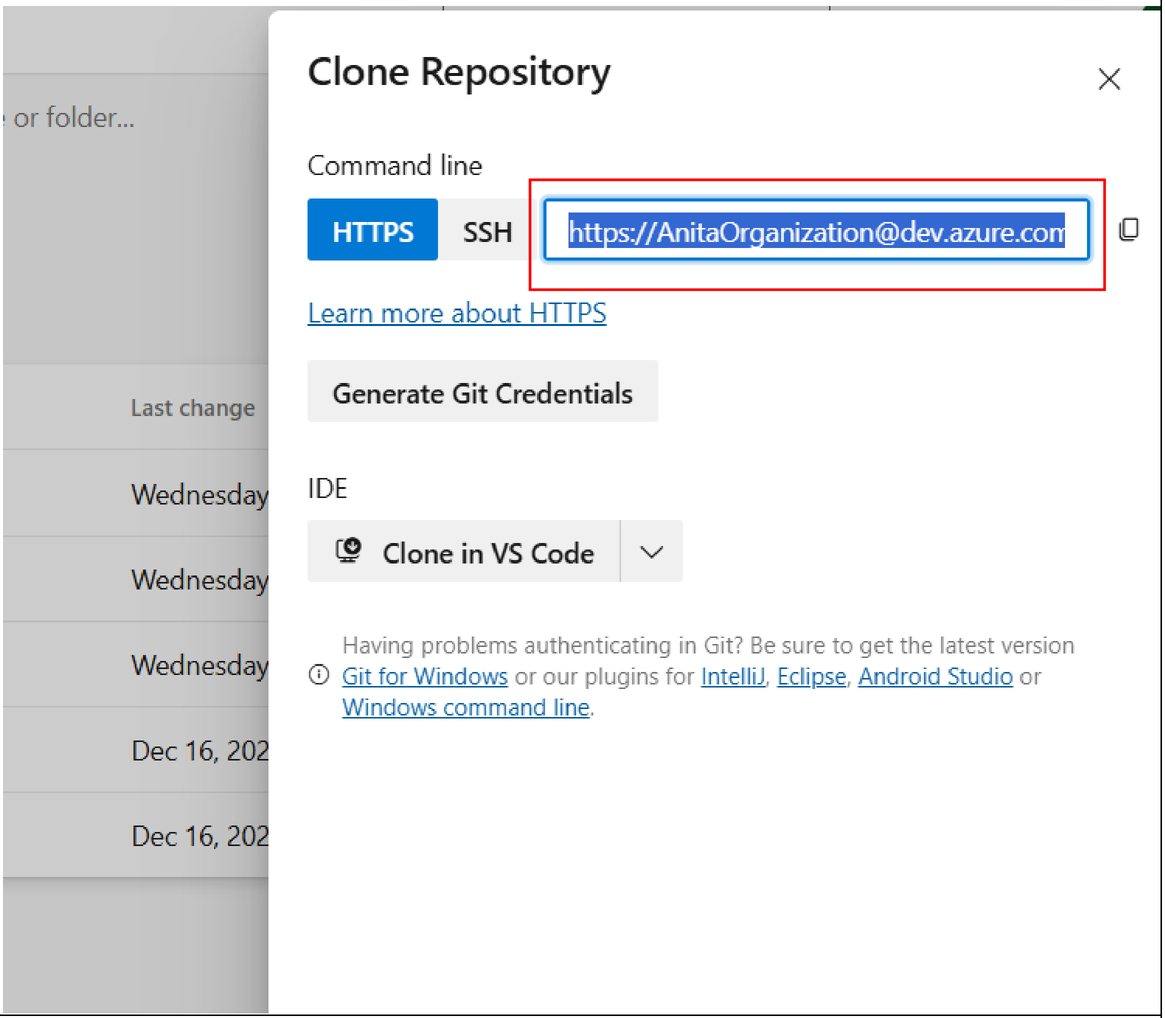

</td></tr>
<tr><td>12</td><td>Now click on dev stage


</td></tr>
</table>

Clone the Repository in Visual Studio

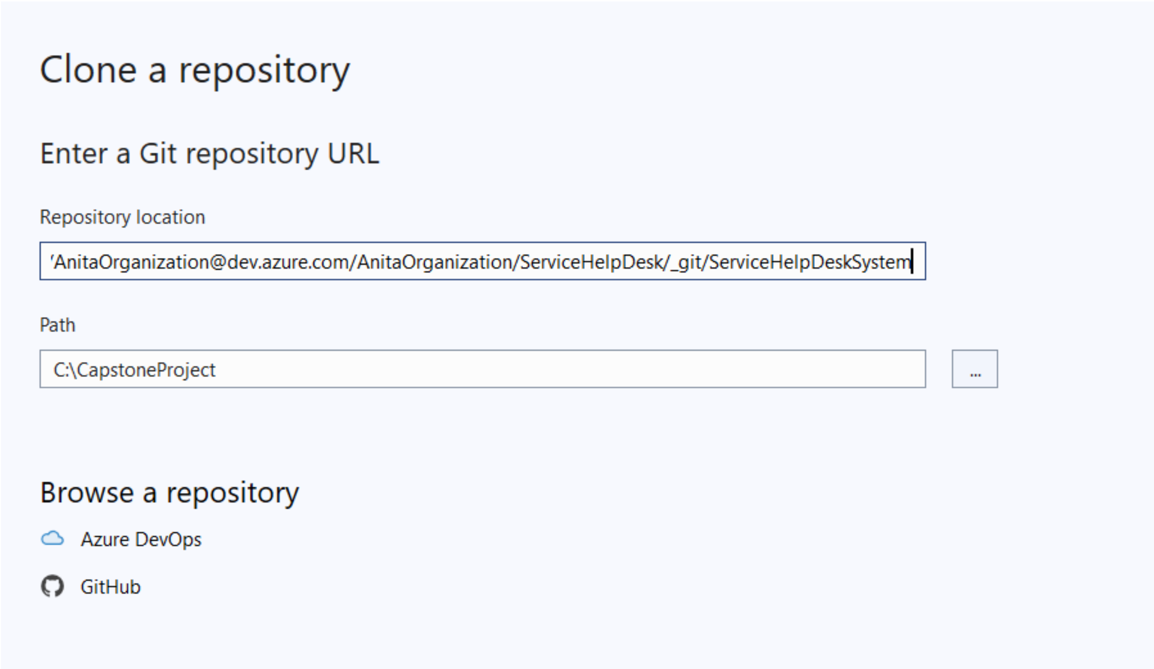
<table>
<tr><td></td><td>Go to Project Repo and Click on the Clone button


</td></tr>
<tr><td></td><td>Copy the URL</td></tr>
</table>

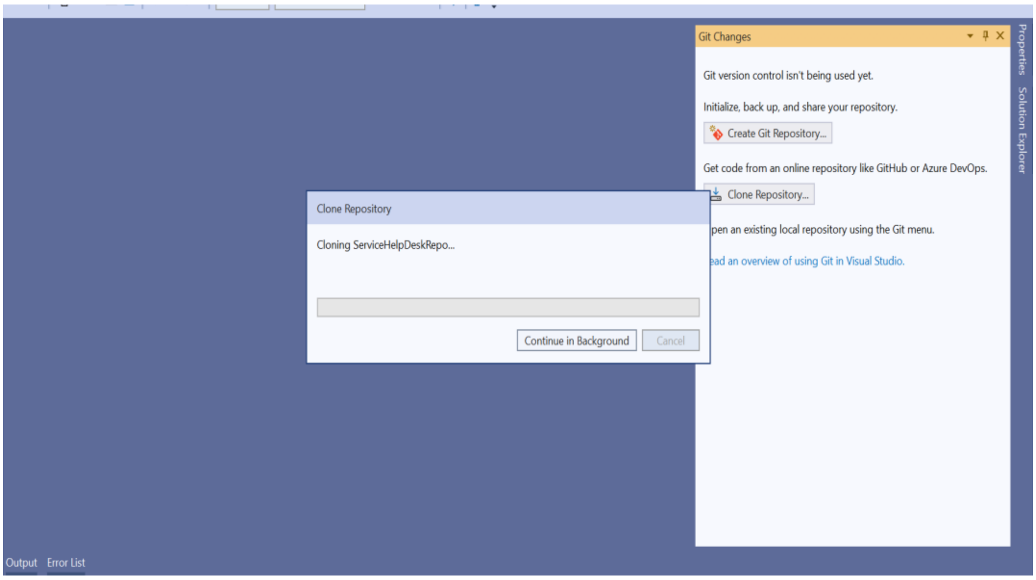


Open Visual Studio click on Clone a repository Option

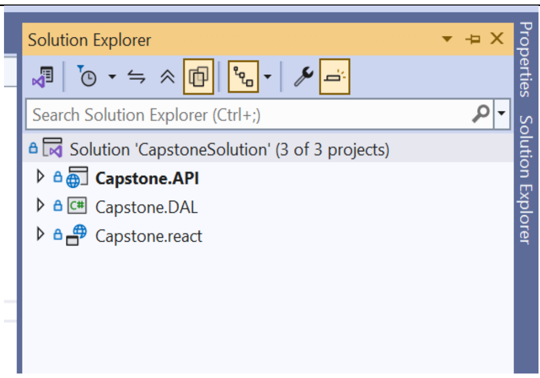
Paste the Clone the URL and Select an Empty Folder , Click on Clone button



This will clone whole repository project on your local path



After Cloning Open the VS solution explorer as below

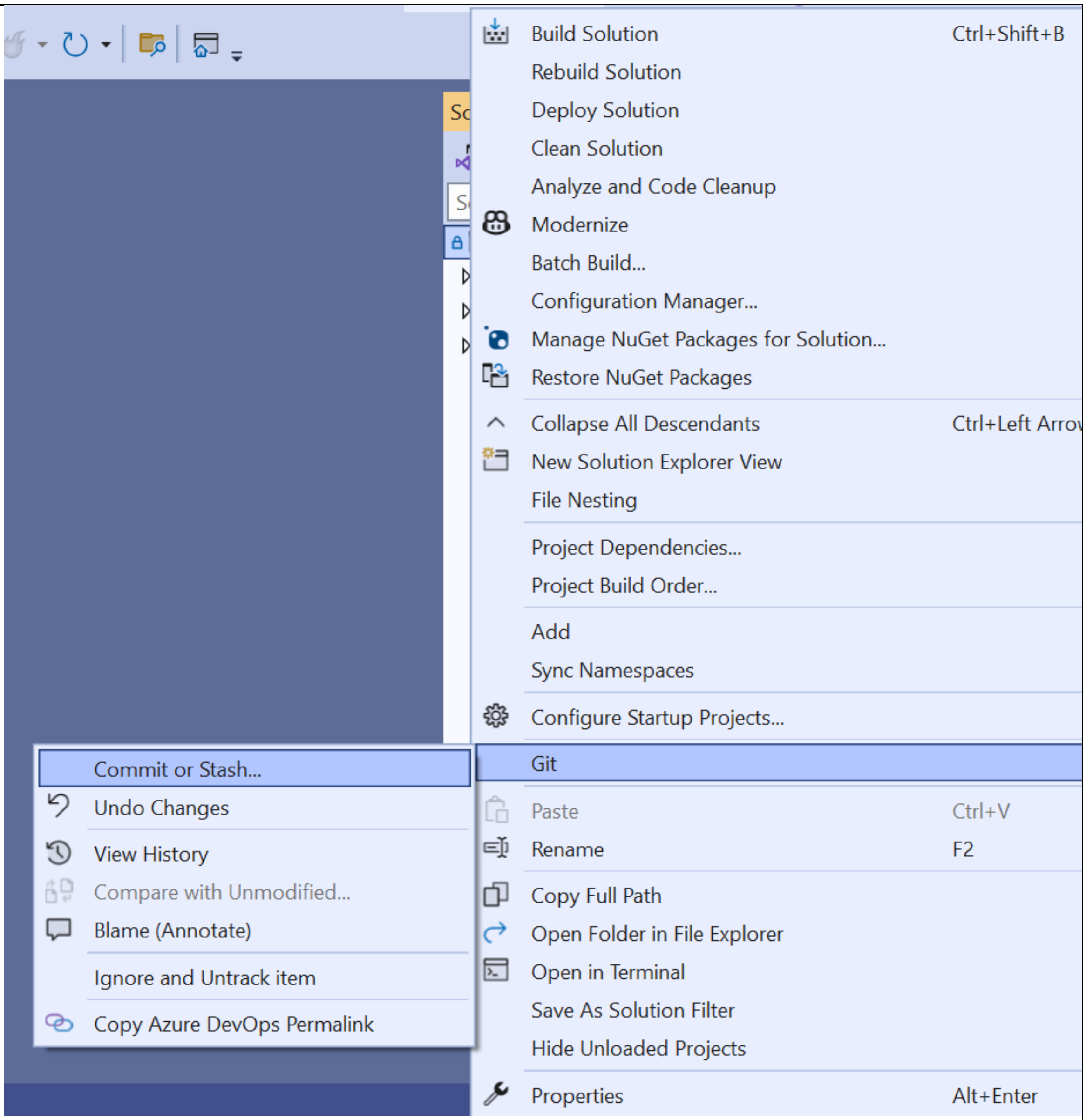


Check the visual studio must be log-in with your Azure credentials.

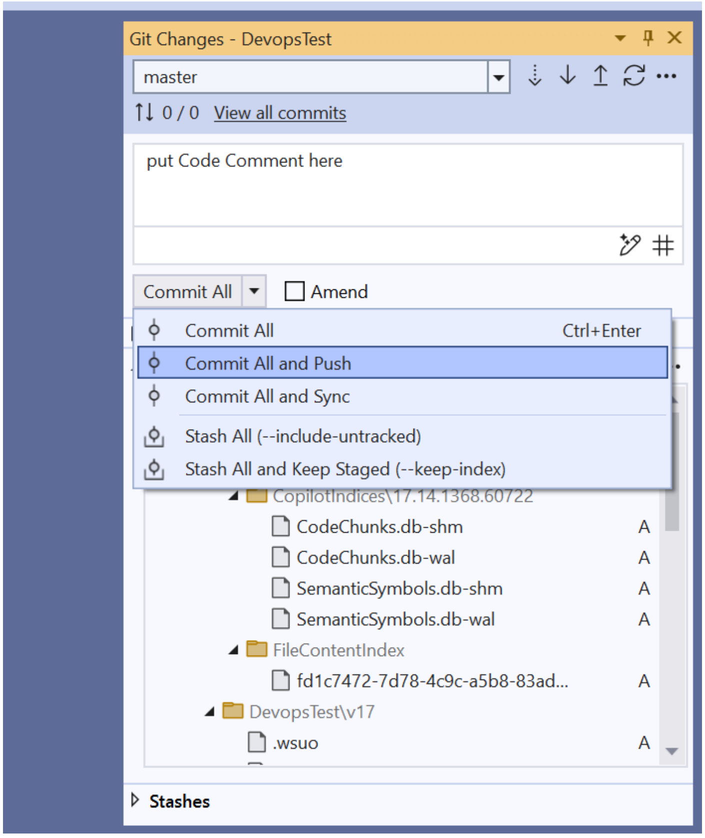
Implement the project specific code in correcponding layers as per Capstone project requirement document .

- DAL
- API
- React presentation

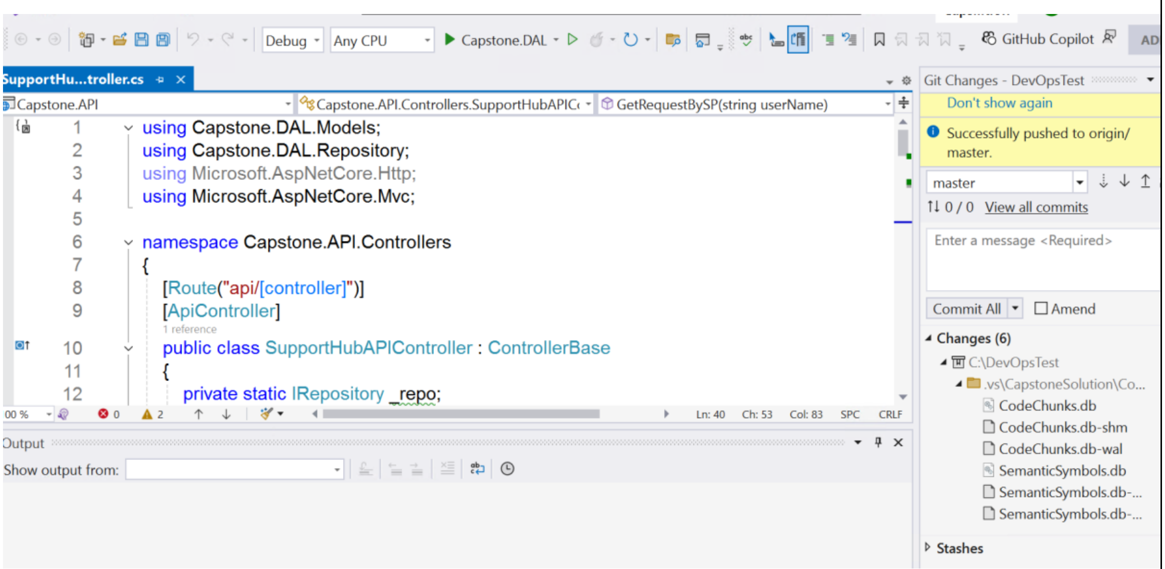
Once the implementation is done, push the changes into central repository by rightclick on Solution explorer, Select Git option


Build Solution Ctrl+Shift+B
Rebuild Solution
Deploy Solution
Clean Solution
Analyze and Code Cleanup
Modernize
Batch Build...
Configuration Manager...
Manage NuGet Packages for Solution...
Restore NuGet Packages
Collapse All Descendants
New Solution Explorer View
File Nesting
Project Dependencies...
Project Build Order...
Add
Sync Namespaces
Configure Startup Projects...
Git
Paste Ctrl+V
Rename F2
Copy Full Path
Open Folder in File Explorer
Open in Terminal
Save As Solution Filter
Hide Unloaded Projects
Properties Alt+Enter
Commit or Stash...
Undo Changes
View History
Compare with Unmodified...
Blame (Annotate)
Ignore and Untrack item
Copy Azure DevOps Permalink

Put the proper comment as per your code and Select the option **Commit All and Push**

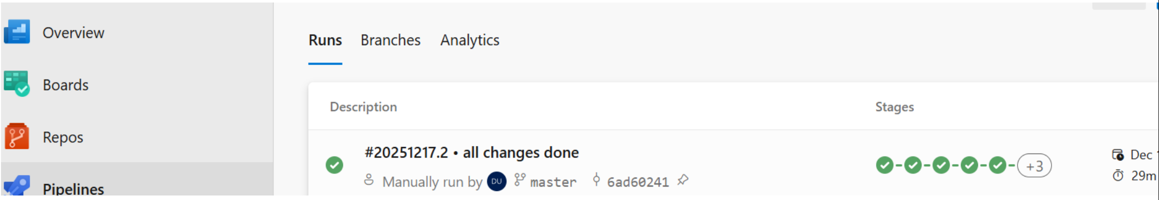


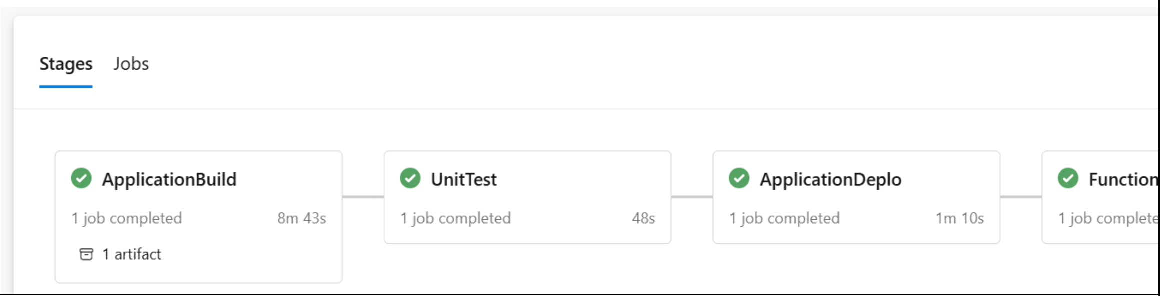
Once changes get pushed into Repo, it shows *successfully pushed to origin* message



Once successful commit is done from visual studio check the project repository in Azure Devops portal. All the new changes will be seen in project Repo.

Go to Project pipeline and Run again. It should successfully run all the stages





APPENDIX (Optional)

To get the Client IP addrees

1. Open the new tab in the browser and go to URL https://www.whatismyip.com/what-is-my-public-ip-address/
2. That will show the address similar to below one. Copy the address , which is the client IP value to be entered.